

Geometry and Topology

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CHAPTER I

Invitation

I.1. The square roots of a complex number

Given a positive real number x , there are two distinct square roots of x , a positive one $\sqrt{x}$ and a negative one $-\sqrt{x}$. Indeed, we have a continuous function $f(x) = \sqrt{x}$, for $x > 0$. If you don't know the definition of a continuous function, you may, for the moment, simply think of it as a function f such that, for any x, y in the domain of x , if y is sufficiently close to x , then $f(x)$ and $f(y)$ can be arbitrarily close to each other.

Now consider a nonzero complex number $z = x + iy$ where $x, y \in \mathbb{R}$. Unlike in the case of real numbers, z always has 2 distinct square roots, and it is certainly tempting to write them as $\pm\sqrt{z}$. But we run into a question: which one of the 2 roots should be $\sqrt{z}$? In the case of positive real numbers, this question has an easy answer: we always take $\sqrt{x}$ to be the positive square root of x . But this answer does not apply to the case of complex numbers, because the two square roots of z are generally not real numbers.

Let us try to find a “coherent way” of assigning a value to $\sqrt{z}$. In other words, we wish to define a continuous function $f : \mathbb{C} \rightarrow \mathbb{C}$ such that $f(z)^2 = z$. For simplicity, we focus on z such that $|z| = 1$. In other words, we let $z \in S$, where S denote the unit circle on the complex plane centered at 0. In this case the image of f is contained in S . Then we have $z = \exp 2\pi i\theta = \cos \theta + i \sin \theta$, where θ is the angle from 1, regarded as the planar vector $(1, 0)$, to z , regarded as the planar vector $(\cos \theta, \sin \theta)$. De Moivre's Theorem says that for any $\theta, \phi \in \mathbb{R}$, we have

$$(I.1.1) \quad \exp 2\pi i\theta \cdot \exp 2\pi i\phi = \exp 2\pi i(\theta + \phi).$$

Now, a reasonable definition of f seems to be $f(\exp 2\pi i\theta) = \exp \pi i\theta$. Of course we'd better fix an interval for the possible values of θ , say $I = [0, 2\pi)$.

We easily verify $f(1) = 1$. On the other hand, if we take $w \in S$ in the fourth quadrant and very close to 1, say $w = \exp 1.998\pi i$, then we have $f(w) = \exp 0.999\pi i$, which is very close to -1 , and quite far away from 1.

At this point it should not be hard to convince yourself that the f thus defined is not continuous at 1. Intuitively speaking, the function f “tears the circle apart” at 1, and then “shrinks” it by half. See Figure 1 for an illustration.

EXERCISE I.1.1. Using the $\epsilon - \delta$ definition of continuity, show that the function f defined above is not continuous at 1.

REMARK I.1.2. One runs into a similar problem when trying to define a continuous logarithm function $g : S \rightarrow \mathbb{R}$ such that $\exp g(z) = z$. Try it yourself.

You may try to fix this by adjusting the interval in which θ takes value, say letting $I = [-\pi, \pi)$. Then the problem is solved “locally” for $\theta = 0$. But then

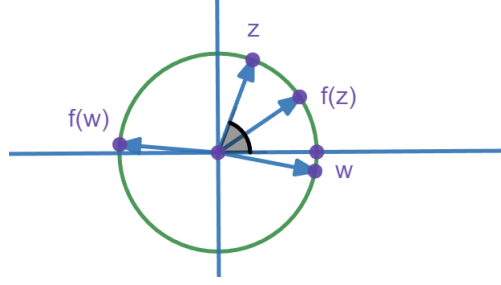


FIGURE 1. the square root of a complex number

you will have to “tear the circle apart” at -1 . On a “global” scale, the problem persists.

You may doubt, then, that a continuous function f that takes the square root of complex numbers does not exist. But it does take some ingenuity to get things started in this direction.

Let us take a step back and consider the function of taking squares: $S \rightarrow S, z \mapsto z^2$. Intuitively, it looks like stretching the unit circle by twice and wind it around another unit circle twice. More generally, for a natural number n the function $z \mapsto z^n$ stretches the unit circle and winds it around another unit circle n times. We look into the winding process more closely in the next section.

I.2. Winding numbers

We would like to study how plane curves wind around a given point. However, at this moment we do not have many tools at our disposal, and it is not quite the right time to consider plane curves in a very general sense. Therefore, we content ourselves with the study of the simplest kind of curves, which we define as follows.

DEFINITION I.2.1. By a *directed planar polyline*, or simply a *polyline* for the rest of this chapter, we mean an ordered collection of directed line segments, $L = (l_1, \dots, l_r)$ where the terminal point of l_i is the initial point of l_{i+1} .

All planar angles in this section are oriented such that counter-clockwise angles are positive.

DEFINITION I.2.2. Let $L = (l_1, \dots, l_r)$ be a polyline. Let the initial and terminal end points of l_i be a_i and a_{i+1} , respectively.

Let p be a point on the plane, and let θ_i be the angle $\angle a_i p a_{i+1}$. Then the sum $\theta = \sum_i \theta_i$ is called the *angle of L around p* .

See Figure 2 for an illustration.

DEFINITION I.2.3. Let $L = (l_1, \dots, l_r)$ be a polyline. We say that L is *closed* if the initial point of l_1 is the terminal point of l_r .

Let θ be the angle of a closed polyline L around a point P . Then the two sides of the angle θ coincide. Therefore we have $\theta = 2\pi n$ for some integer n .

REMARK I.2.4. The “closeness” condition on L leads to a strong restriction on the value of its angle around a point. In Section I.3 we will see the analog of this phenomena in 3 dimensions.

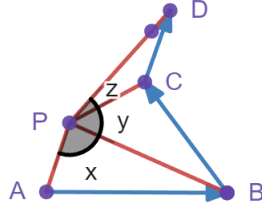


FIGURE 2. The angle of the polyline $(\vec{AB}, \vec{BC}, \vec{CD})$ around P is $x + y + z$.

DEFINITION I.2.5. Let L be a close polyline and $p \in L$ a point on the same plane. The *winding number* of L around p is the integer n such that the angle of L around p is $2\pi n$. We denote this winding number by $W(L, p)$.

EXAMPLE I.2.6. Let $L = (l_1, \dots, l_r)$ be the successive edges of a polygon C , each l_i assigned a direction such that they collectively form a counter clockwise loop. Then L is a closed polyline. See Figure 3 for an illustration.

Let p be a point inside C , you will likely be convinced, after working on several examples, that we have $W(L, p) = 1$.

Let $L^2 = (l_1, \dots, l_r, l_1, \dots, l_r)$, then you may be convinced of $W(L^2, p) = 2$.

If each l_i is given the opposite direction, then we have $W(L, p) = -1$.

Let q be a point outside C , then you will likely be convinced of $W(L, q) = 0$.

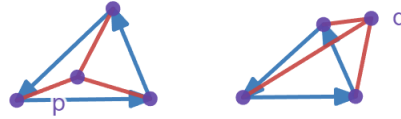


FIGURE 3. The winding number of the edges of a polygon, around a point inside or outside of it.

For the moment we will not attempt to prove the statements in Example I.2.6. Although you might try to present one by yourself.

We would certainly like to have a definition of winding numbers for planar loops of more general forms rather than merely the closed polylines. For instance, we would like to regard the function

$$\sigma : [0, 1] \rightarrow S, \quad t \mapsto \exp 2\pi i n t, \quad n \in \mathbb{Z}$$

as a legit planar loop. Intuitively, this function winds the interval $[0, 1]$ around the circle S for n times, and we would like a definition of winding number, generalizing our existing one, and applies to σ , such that the winding number of σ around any point inside of S .

At this point, you might suspect that the notion of winding numbers is somehow related to the problem of finding a continuous square root function in Section I.1, and you would be right. This will be the topic of Chapter III.

I.3. Angel defect, the Gauss-Bonnet formula for polyhedra

In this section, we consider polyhedra, the natural generalization of polygons to the 3 dimensional space.

DEFINITION I.3.1. A *polyhedron* P is a collection of distinct polygons $F_1, \dots, F_r$ in the 3 dimensional space $\mathbb{R}^3$, called the *faces* of P , subjected to the following conditions:

- (P1) If $i \neq j$, then $F_i \cap F_j$ is either $\emptyset$ or a common edge of F_i and F_j .
- (P2) Let E, E' be edges of some faces of P , and assume $E \neq E'$. Then $E \cap E'$ is either $\emptyset$ or a vertex of one of the faces.
- (P3) For each edge E , there are exactly two faces F, F' such that $E = F \cap F'$.

The edges (or vertices) of the faces of P are called the *edges* (or *vertices*) of P .

A polyhedron is among the simplest examples of “closed surfaces” in $\mathbb{R}^3$. We proceed to define a quantity measures how a polyhedron is “bent” at a vertex.

Let P be a polyhedron, $F_1, \dots, F_r$ distinct faces of P such that $F_i \cup F_j = E_{ij}$ are distinct edges, and that $V = E_1 \cap E_2$ is a vertex of P . Let α_i be the inner angle of F_i at V .

DEFINITION I.3.2. The *angle defect* of P at V is $2\pi - \sum_{i=1}^r \alpha_i$. The sum of the angle defects of all vertices of P is called the *total angle defect* of P and is denoted by $\sigma(P)$.

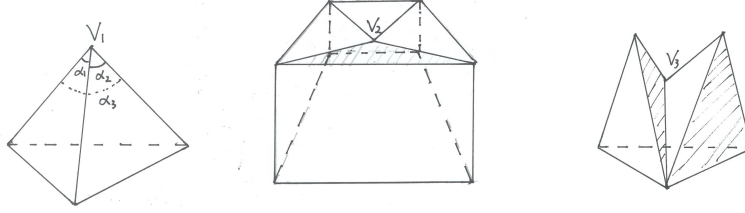


FIGURE 4. Typical cases of angle defects.

In Figure 4, the angle defect at V_1 is $2\pi - \sum_{i=1}^3 \alpha_i$. If you carefully assign appropriate values to the angles in each polyhedron in Figure 4, you will find that the angle defect at V_1 and V_2 positive, whereas at V_3 , the angle defect is negative.

Let us look at the two cases where the angle defect is positive. The landscape near V_1 looks like the top of a mountain, and that near V_2 looks like the bottom of a basin. What they have in common can be Intuitively describe as follows: If you are to travel on the polyhedron from V_1 or V_2 , then either all directions are ascendent, or all of them are descendent.

Now consider the case where the angle defect is negative. The landscape near V_3 looks like the source of a creek: From V_3 , you can either ascend or descend.

EXERCISE I.3.3. Compute the total angle defect of a regular tetrahedron and a cube.

In both case, you will find that the correct answer is 4π . Indeed, if you assign proper values to the inner angles of all faces of the polyhedra in FIGure 4 and compute the total angle defects of all three of them, you will obtain the same value.

For mathematicians, this is a good reason to start looking for more things that the aforementioned polyhedra have in common. There is one observation we can make, although it is not mathematically rigorous. Imagine that the aforementioned polyhedra, are made of elastic membrane. If we take any of them and fill it with air, it is to be “blown up” and deform into a sphere.

We proceed to show that this observation is relevant to our computation of angle defects. Consider the polyhedron T in Figure 5. If you have enough patient, you can show that the total angle defect of T is $\sigma(T) = 0$, which is different from all the examples seen earlier. And if you are to “blow up” this polyhedron, you get a shape which looks like the surface of a tire, or a donut, which is visually distinguishable from a sphere. Mathematicians call this shape a torus. Intuitively, a torus and a sphere cannot “deform” into each other. This sentence can be made into a precise and correct mathematical statement, although it would be premature to discuss it now.

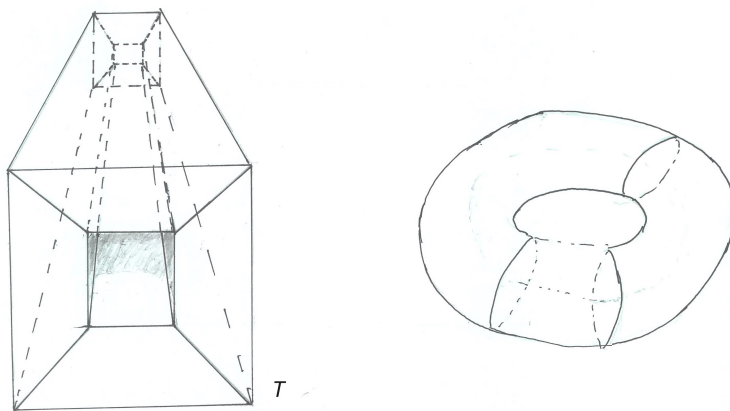


FIGURE 5. The polyhedron T can be “deformed into” a torus.

We proceed to consider another quantity associated to a polyhedron.

DEFINITION I.3.4. Let P be a polyhedron. Let v, e, f be the total numbers of the vertices, edges, and faces of P , respectively. The *Euler characteristic* of P , denoted $\chi(P)$, is defined as

$$\chi(P) := v - e + f.$$

EXERCISE I.3.5. Find the Euler characteristics of the cubic and the polyhedra in Figure 4.

EXERCISE I.3.6. Find $\chi(T)$.

After all the hard work, a nice pattern emerges, and it is known in the form of the following

THEOREM I.3.7. *Let P be a polyhedron. We have $\sigma(P) = 2\pi\chi(P)$.*

An elegant proof can be found in Givental’s notes [Giv].

□

REMARK I.3.8. The quantities $\sigma(P)$ and $2\pi\chi(P)$ may first appear to you as of very different nature: the value of $\sigma(P)$ appears to be arbitrary, as the angle defect at one vertex may run over an interval as the P gets “deformed near V ”. It is the sum of “locally” defined quantities.

The quantity $2\pi\chi(P)$, on the other hand, is obviously an integral multiple since $\chi(P)$ is an integer by definition. There is no obvious way to decompose $\chi(P)$ into a sum of “locally” defined quantities.

There are many deep theorems in mathematics, especially in geometry and topology, that unifies two quantities about one geometric object, one of which is continuous and local, whereas the other is discrete and global.

Based on the examples considered in this section, we may conjecture that if two polyhedra P and Q can be “deformed into the same shape”, which can be a sphere or a torus, then we have $\sigma(P) = \sigma(Q)$, or equivalently $\chi(P) = \chi(Q)$. Then, is it possible to define the “total angle defect”, or “Euler characteristic” of a sphere or a torus, without any explicit reference to a particular polyhedron? The answer is yes, and this would require a considerable amount of knowledge in modern geometry and topology, which will be taught in the rest of this course.

CHAPTER II

General Topology

II.1. Continuous mappings between Euclidean spaces

DEFINITION II.1.1. Let U be a subset of $\mathbb{R}^m$ and $f: U \rightarrow \mathbb{R}^n$ a mapping. Then f is *continuous at x* if for any $\epsilon > 0$, there is a $\delta > 0$, such that for each $y \in U$ satisfying $|y - x| < \delta$, there is $|f(y) - f(x)| < \epsilon$. We simply say that f is continuous if f is continuous at all points $x \in U$.

We say f is *discontinuous at x* if it is not continuous at x .

REMARK II.1.2. In general, the choice of δ depends on x and ϵ .

EXAMPLE II.1.3. Let $f: \mathbb{R}^m \rightarrow \mathbb{R}^n$ be the constant function taking value $c \in \mathbb{R}^n$. Then f is continuous at every $x \in \mathbb{R}^m$.

PROOF. For any $\epsilon > 0$, take $y \in \mathbb{R}^m$ such that $|y - x| < 1$ (or replace 1 with any positive δ you like), there is

$$|f(y) - f(x)| = |c - c| = 0 < \epsilon.$$

□

EXAMPLE II.1.4. Let $k \in \mathbb{R}$. Consider $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$, $f(x) = kx$. Then $f(x)$ is continuous at every $x \in \mathbb{R}^n$.

PROOF. If $k = 0$, then f is a constant function, and the conclusion follows from Example II.1.3.

Otherwise, take $\epsilon > 0$. For any $y \in \mathbb{R}^n$ satisfying $|y - x| < \epsilon/|k|$, we have

$$|f(y) - f(x)| = |k||y - x| < |k| \frac{\epsilon}{|k|} = \epsilon.$$

□

In the previous two examples, the choice of δ is independent of x . This is not the case for the next example.

EXAMPLE II.1.5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$. Then f is continuous at every $x \in \mathbb{R}$.

PROOF. For $\epsilon > 0$, let

$$0 < \delta < \min\left\{1, \frac{\epsilon}{1 + 2|x|}\right\}.$$

Then for $y \in \mathbb{R}$, $|y - x| < \delta$, we have

$$|y| < |x| + 1,$$

and

$$\begin{aligned}
|f(y) - f(x)| &= |y - x||y + x| \\
&\leq |y - x|(|y| + |x|) \\
&\leq |y - x|(1 + |x| + |x|) \\
&< \frac{\epsilon}{1 + 2|x|}(1 + 2|x|) = \epsilon.
\end{aligned}$$

□

REMARK II.1.6. A direct way to say that $f: U \rightarrow \mathbb{R}^n$ is discontinuous at $x \in U$ is as follows: There exist an $\epsilon > 0$, such that for any $\delta > 0$ there is a $y \in U$, satisfying $|y - x| < \delta$ and $|f(y) - f(x)| \geq \epsilon$.

EXAMPLE II.1.7. Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} -1, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then f is discontinuous at 0.

PROOF. Fix your favorite $0 < \epsilon < 2$. For any $\delta > 0$, let $y = -\delta/2$. Then we have $|y - 0| < \delta$ and $|f(y) - f(0)| = 2 > \epsilon$. □

II.2. Metric Spaces and Continuous Maps

Given an Euclidean space $\mathbb{R}^n$, and two points

$$x = (x_1, \dots, x_n), y = (y_1, \dots, y_n), \in \mathbb{R}^n$$

the *distance* between x, y is defined as

$$(II.2.1) \quad \rho(x, y) := \left[\sum_{i=1}^n (x_i - y_i)^2 \right]^{1/2}.$$

The function $\rho(-, -)$ satisfies three properties:

- D1 $\rho(x, y) \geq 0$, and the equality holds if and only if $x = y$;
- D2 $\rho(x, y) = \rho(y, x)$;
- D3 $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$.

Sometimes we are interested in the function $\rho: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ without caring much about the concrete defining formula (II.2.1). This motivates the following

DEFINITION II.2.1. A *metric* on a set X is a function $\rho: X \times X \rightarrow \mathbb{R}_{\geq 0}$ satisfying D1 - D3. The pair (X, ρ) is called a *metric space*. When no ambiguity is possible, we simply say that X is a metric space.

EXERCISE II.2.2. Let $f: X \rightarrow Y$ be a mapping between metric spaces. Formulate a definition of f being continuous at $x \in X$. In the case where $X \subset \mathbb{R}^m$ and $Y = \mathbb{R}^n$, your definition should be consistent with the definition of continuity given earlier.

We introduce more terminologies on metric spaces motivated by the special case of Euclidean spaces.

DEFINITION II.2.3. Let X be a metric space and let $x \in X$. Let $r > 0$. The *open ball* and the *closed ball* centered at x with radius r are defined respectively as

$$\mathbf{B}(x, r) := \{y \in X : \rho(x, y) < r\}$$

and

$$\bar{\mathbf{B}}(x, r) := \{y \in X : \rho(x, y) \leq r\}.$$

DEFINITION II.2.4. Let X be a metric space. An *open set* in X is a union of open balls. A *closed set* in X is the complement of an open set.

REMARK II.2.5. The empty set and X itself are both open and closed.

PROPOSITION II.2.6. *Let X be a metric space. A subset U of X is an open set if and only if for each $x \in U$, there is an open ball $\mathbf{B}(x, r)$ which is a subset of U .*

The proof is left as an exercise. $\square$

Recall that the closure of a subset A of $\mathbb{R}^n$ is the collection of all limit points of A . We generalize this concept in the context of metric spaces.

DEFINITION II.2.7. Let X be a metric space and A be a subset of X . A *limit point* of A is a point $x \in X$ such that, for any open set U containing x , there is

$$(U - \{x\}) \cap A \neq \emptyset.$$

The *closure* of A , denoted by $\bar{A}$, is the collection of all points in A and all limit points of A .

PROPOSITION II.2.8. (1) *Let X be a metric space and A be a subset of X . Then $\bar{A}$ is a closed set.*
 (2) *Let V be a closed set in X . Then we have $\bar{V} = V$.*

The proof is left as an exercise. $\square$

II.3. Topological spaces

In Section II.2, we defined the open sets in a metric space X . Let τ be the collection of all open sets of X . Then the following properties hold:

T1 Let I be an index set, and $\{U_i\}_{i \in I}$ a subset of τ , then we have $\bigcup_{i \in I} U_i \in \tau$.

T2 If $U, V \in \tau$, then we have $U \cap V \in \tau$.

T3 $\emptyset, X \in \tau$.

It turns out that even with only these properties, a lot can be done.

DEFINITION II.3.1. Let X be a set, and τ a collection of subsets of X satisfying T1 - T3. Then τ is called a *topology* on X . A member of τ is called an *open set* of X . The complement of an open set is called a *closed set* of X . The pair (X, τ) is called a *topological space*. When no ambiguity is possible, we simply call X a topological space.

For $x \in X$, an open set of X containing x is called a *neighborhood* of x .

EXERCISE II.3.2. Let X be a set. Let ξ be a set of subsets of X . Let

$$\tau = \{X - V \mid V \in \xi\}.$$

Show that τ is a topology on X if and only if the following holds:

C1 Let I be an index set, and $\{V_i\}_{i \in I}$ a subset of ξ , then we have $\bigcap_{i \in I} V_i \in \xi$.

C2 If $V, W \in \xi$, then we have $V \cup W \in \xi$.

C3 $\emptyset, X \in \xi$.

In this case, obviously, ξ is the set of all the closed sets in (X, τ) .

EXERCISE II.3.3. Find a set of open sets in $\mathbb{R}$ such that their intersection is not an open set.

The following is simply an reiteration of the first paragraph of this section.

PROPOSITION II.3.4. *If X is a metric space, then the metric determines a collection of open sets τ which is a topology on X .* $\square$

In metric spaces, we have the notions of limit points and closure of a subset, which can be easily generalized in the case of topological spaces.

DEFINITION II.3.5. Let X be a topological space, and let $A \subset X$. A *limit point* of A in X is a point $x \in X$ such that for any neighborhood U of x , the set $U \cap A$ contains some points other than x .

The *closure* of A in X , denoted by $\bar{A}$ is subset of X consisting of all points of A and all limit points of A .

PROPOSITION II.3.6. *Let X and A be as in Definition II.3.5. Then $\bar{A}$ is the intersection of all closed sets containing A as a subset. In particular, $\bar{A}$ is a closed set, and if A is closed, then $\bar{A} = A$.*

PROOF. Let Φ be the collection of all closed subsets containing A . Let $\Gamma = \bigcap_{V \in \Phi} V$ be the union of all members of Φ .

First we show that $\bar{A} \subset \Gamma$. This is the same as showing $\bar{A} \subset V$ for every $V \in \Phi$. Since $X - V$ is an open set, and $(X - V) \cap A = \emptyset$, any point $x \in X - V$ is not a limit point of A . In other words, any limit point of A is in V . Therefore, we have $\bar{A} \subset V$.

To show $\Gamma \subset \bar{A}$ is the same as showing that for any $x \in X - \bar{A}$, there is a $V \in \Phi$ such that $x \notin V$. Indeed, $x \in X - \bar{A}$ means that there is an open set U such that $U \cap A = \emptyset$ and $x \in U$. Taking $V = X - U$, we have $V \in \Phi$ and $x \notin V$. $\square$

DEFINITION II.3.7. Let X, Y be topological spaces. Let $x \in X$. A mapping $f : X \rightarrow Y$ is *continuous* at x if for any open set V in Y such that $f(x) \in V$, there is an open set U in X satisfying $x \in U$ and $U \subset f^{-1}(V)$. When f is continuous at every $x \in X$, we simply say that f is continuous.

In this course, a continuous mapping is also called a *map*.

EXERCISE II.3.8. Show that for a mapping $f : X \rightarrow Y$ where X is a subset of $\mathbb{R}^m$ and $Y = \mathbb{R}^n$, the two definitions of continuity, Definition II.3.7 and Definition II.1.1, agree.

PROPOSITION II.3.9. *A mapping $f : X \rightarrow Y$ between topological spaces is continuous iff the following is true: if V is an open set in Y , then $f^{-1}(V)$ is an open set in X .*

EXERCISE II.3.10. Prove Proposition II.3.9.

The following proposition, in fancier language, says that topological spaces and continuous maps form a “category”.

PROPOSITION II.3.11. (1) *For a topological space X , the identity function $\mathbb{I}_X$ is continuous.*

- (2) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be mappings between topological spaces. Let $x \in X$. If f is continuous at x and g is continuous at $f(x)$, then gf is continuous at x . In particular, if f and g are continuous, then so is $gf : X \rightarrow Z$.

EXERCISE II.3.12. Prove Proposition II.3.11.

The concept of topology, or topological spaces, seems very abstract, even confusing at the first glimpse. However, it plays a fundamental role in modern mathematics, and we will have a taste of its significance as the course unfold.

DEFINITION II.3.13. Let X, Y be topological spaces and $f : X \rightarrow Y$ be a map. Then f is a *homeomorphism* if f is a bijection and f^{-1} is continuous.

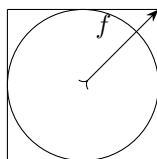
EXAMPLE II.3.14. Let

$$X = \{(x, y) \in \mathbb{R}^2 : -1 \leq x, y \leq 1\}$$

be a square in $\mathbb{R}^2$, and let

$$Y = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$$

be a disk. Both X and Y inherit the distance function from $\mathbb{R}^2$. Therefore they are metric spaces, whence topological spaces. We have a map $f : X \rightarrow Y$, which sends a point $(x, y) \in X$ to the unique point of Y of the form (tx, ty) where $t > 0$, as indicated by the picture (II.3.14).



(II.3.1)

It is not hard to check that f is a homeomorphism. This example is an illustration that topology concerns the properties of geometric objects that are invariant with respect to “stretching and shrinking”. For this reason, sometimes topology is nicknamed “rubber sheet geometry”.

REMARK II.3.15. Following Example II.3.14, we may ask a further question: Let $X' := X - (0, 0)$ be the “punctured disk”, then is there a homeomorphism $X \rightarrow X'$? It turns out that trying very hard to find a such a homeomorphism does not help. The question is answered, in a rather indirect way, in Chapter III.

A set may be equipped with more than one topology. Sometimes we can compare two topologies on the same set.

DEFINITION II.3.16. Let X be a set, and $\tau_i, i = 1, 2$ be two topologies on X . If $\tau_1 \subset \tau_2$, then we say that the topology τ_1 is *coarser* than τ_2 , or τ_2 is *finer* than τ_1 .

EXAMPLE II.3.17. A set X always have two special topologies: the “coarsest” one, the *trivial topology*, is simply $\{X, \emptyset\}$, and the “finest” one, the *discrete topology*, consisting of all the subsets of X .

EXERCISE II.3.18. Let X be a set, Y a topological space. Show that

- (1) if X is equipped with the discrete topology, then any function $f : X \rightarrow Y$ is continuous;

- (2) if X is equipped with the trivial topology, then any function $g : Y \rightarrow X$ is continuous.
- (3) Find two topological spaces X, Y and a continuous bijection $f : X \rightarrow Y$ such that f^{-1} is not continuous.

Let us take a look at a less extreme example.

EXERCISE II.3.19. Consider $\mathbb{R}$, the set of real numbers. Let

$$\tau := \{\mathbb{R} - F : F \text{ is a finite subset.}\} \cup \{\emptyset\}.$$

Show that τ is a topology on $\mathbb{R}$, thereby a coarser topology than the usual one.

Exercise II.3.19 is an instance of a very important kind of topology:

DEFINITION II.3.20. Let $\mathbb{F}$ be a field. (If you don't know what a field yet, then think of $\mathbb{F}$ as one of $\mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$.) Let n be a positive integer. A *Zariski closed set* in the n -fold Cartesian product $\mathbb{F}^n$ is of the form

$$V := \{(x_1, \dots, x_n) \in \mathbb{F}^n : f_i(x_1, \dots, x_n) = 0, \quad i \in I\}$$

where I is an index set and f_i 's are polynomials in n variables with coefficients in $\mathbb{F}$. Then the *Zariski topology* τ_Z on the n -fold Cartesian product $\mathbb{F}^n$ is defined by

$$\tau_Z := \{\mathbb{F}^n - V : V \text{ is a Zariski closed set.}\}.$$

A member of τ_Z is called a *Zariski open set* in $\mathbb{F}^n$.

- EXERCISE II.3.21. (1) Show that τ_Z is indeed a topology on $\mathbb{F}^n$.
 (2) Show that the topology on $\mathbb{R}$ defined in Exercise II.3.19 is the Zariski topology on $\mathbb{R}$.

The notion of Zariski topology plays a fundamental role in a branch of mathematics called algebraic geometry.

Sometimes it can be difficult to describe all the open sets of a topological space. The following terminology may help.

DEFINITION II.3.22. Given a topological space (X, τ) , a *basis* of (X, τ) is a subset $B \subset \tau$ such that any $U \in \tau$ is a union of members in B . In this case we say that τ is *generated by* B .

EXAMPLE II.3.23. The set of open balls in a metric space X form a basis of the topology on X associated to the metric. This follows immediately from the definition of the open sets in X .

DEFINITION II.3.24. Let X be a set and B a set of subsets of X . Then B is said to have the *basis properties* if the following holds:

- $X = \bigcup_{U \in B} U$, and
- if $U, V \in B$, then for each point $x \in U \cap V$, there is a $W \in B$ such that $x \in W$ and $W \subset U \cap V$.

PROPOSITION II.3.25. *Let X and B be as above. Then B is a basis for a topology on X if and only if B has the basis properties.*

EXERCISE II.3.26. Prove Proposition II.3.25.

The proof is left to the reader an exercise. □

EXERCISE II.3.27. Let X be a set and let ρ, τ be metrics on X . If there are $a, b > 0$ such that for any $x, y \in X$, we have

$$a\rho(x, y) < \tau(x, y) < b\rho(x, y).$$

Show that ρ and τ defines the same topology on X .

(In this case we say that f is a *uniform equivalence* of the metrics ρ and τ .)

II.4. Deriving new topologies from existing ones

Given a collection of sets, there are various ways to derive new sets out of them. For instance, one may take the subset of a set, the quotient set of a set modulo an equivalence relation, the product/disjoint union of a collection of sets, and so on. Similar things can be done to a given collection of groups. For topological space, this holds true as well.

EXERCISE II.4.1. Let (X, τ) be a topological space and A a subset of X . Then the set

$$\tau_A = \{U \cap A : U \in \tau\}$$

is a topology on A .

DEFINITION II.4.2. Taking the notations above, (A, τ_A) is called the *subspace* of X with underlying set A . We often simply say that A is a subspace of X . We call τ_A the *subspace topology* on A .

EXERCISE II.4.3. If X is a metric space, show that the topology on subspace A induced by the restriction of the distance function agrees with the subspace topology on A .

EXERCISE II.4.4. Let $I = [0, 1]$, a closed interval in $\mathbb{R}$. Let $\mathbb{R}$ have the usual topology, which is the topology associated to its metric $\rho(x, y) = |x - y|$. Let I have the subspace topology. Show that $[0, \frac{1}{2})$ is not an open set of $\mathbb{R}$, and it is an open set of I .

When we consider “geometric objects” such curves and surfaces (whatever they means) in Euclidean spaces, we most often regard them as subspaces of Euclidean spaces.

If we regard taking a subset as an “operation” on a set that produces another set, then what we have done is upgrading this operation to an “operation” on topological spaces. In the rest of this section, we do this for more complicated “operations” on sets.

Given a collection of sets $\{X_i\}_{i \in I}$ where I is an index set, we have the disjoint union

$$(II.4.1) \quad \sqcup_{i \in I} X_i := \{(i, x_i) : i \in I, x_i \in X_i\}.$$

For each i there is a canonical function

$$(II.4.2) \quad \iota_i : X_i \rightarrow \sqcup_{i \in I} X_i, x_i \mapsto (i, x_i)$$

which is injective. Therefore, each X_i may be regarded as a subset of $\sqcup_{i \in I} X_i$.

We also have the Cartesian product

$$(II.4.3) \quad \prod_{i \in I} X_i := \{f : I \rightarrow \sqcup_{i \in I} X_i : f(i) \in X_i\}.$$

Again, for each i there is a canonical function

$$(II.4.4) \quad \prod_{i \in I} X_i \rightarrow X_i, f \mapsto f(i)$$

which is surjective. A more common convention is to write $f(i)$ as something like x_i and f as $(x_i)_{i \in I}$ as we do for $\mathbb{R}^n$, the n -fold Cartesian product of $\mathbb{R}$.

Let I be an index set. For each $i \in I$, let (X_i, τ_i) be a topological space. Recall the definition of a basis of a topology, Definition II.3.22.

EXERCISE II.4.5. Let ι_i be as in (II.4.2). Show that the set $\bigcup_{i \in I} \iota_i(\tau_i)$ satisfies the basis properties.

DEFINITION II.4.6. Taking the notations above, the *disjoint union topology* of $\sqcup_{i \in I} X_i$ is the topology generated by $\bigcup_{i \in I} \iota_i(\tau_i)$.

We may think of, for instance, the disjoint union of two discs as the subspace of $\mathbb{R}^2$ consisting of two discs with no intersection, although the “ambient space” $\mathbb{R}^2$ is irrelevant.

Next, we consider the Cartesian products. Let I be an index set and for $i \in I$, let (X_i, τ_i) be a topological space.

EXERCISE II.4.7. Show that the set of subsets of $\prod_{i \in I} X_i$ defined by

$$B = \left\{ \prod_{i \in I} U_i : U_i \in \tau_i, U_i = X_i \text{ for all but finitely many } i \in I. \right\}$$

and

$$B' = \left\{ \prod_{i \in I} U_i : U_i \in \tau_i, i \in I. \right\}$$

have the basis properties.

DEFINITION II.4.8. • The topology on $\prod_{i \in I} X_i$ generated by B is called the *product topology* of $\prod_{i \in I} X_i$.

• The topology on $\prod_{i \in I} X_i$ generated by B' is called the *box topology* of $\prod_{i \in I} X_i$.

Notice that the two topologies agree when the index set I is finite.

EXERCISE II.4.9. Let τ denote the product topology on $\prod_{i \in I} X_i$. Let τ' be a topology on $\prod_{i \in I} X_i$. Then $\tau \subset \tau'$ if and only if for all $j \in I$, the projection map

$$p_j : \left(\prod_{i \in I} X_i, \tau' \right) \rightarrow (X_j, \tau_j)$$

is continuous.

EXERCISE II.4.10. Consider $\mathbb{R}$ with its usual topology. Show that the product topology of $\mathbb{R}^n$ agrees with the topology on $\mathbb{R}^n$ induced by the Euclidean metric.

EXERCISE II.4.11. Let $\mathbb{R}^{\mathbb{N}}$ be the Cartesian product $\prod_{i \in \mathbb{N}} \mathbb{R}_i$ where $\mathbb{R}_i = \mathbb{R}$. Consider the function

$$d : \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}, x \mapsto (x, x, \dots).$$

If $\mathbb{R}$ is given the usual topology, then is d continuous when $\mathbb{R}^{\mathbb{N}}$ is given the product topology/box topology?

Given a set X and a subset R of $X \times X$. We write $x \sim y$ for $(x, y) \in R$. We call R an *equivalence relation* on X if it satisfies

- (1) $x \sim x$ (reflexivity);
- (2) if $x \sim y$, then $y \sim x$ (symmetry);
- (3) if $x \sim y$, $y \sim z$, then $x \sim z$ (transitivity).

Given a set X with an equivalence relation R on X , an *equivalence class* in X under R is a subset A of X such that for any $a \in A$, $x \in X$, we have $a \sim x$ if and only if $x \in A$. In other words, an equivalence class is a subset of X such that any two points in it are equivalent, and any point outside of it is not equivalent to any point in it.

EXERCISE II.4.12. Let A, B be equivalence classes in X . Show that either $A = B$ or $A \cap B = \emptyset$.

EXERCISE II.4.13. Show that X is the union of the equivalence classes.

DEFINITION II.4.14. Given a set X and an equivalence relation R on X . The *quotient set* of X under R , denoted as X/R , read as “ X modulo R ”, is the set of equivalence classes in X under R .

The function $X \rightarrow X/R$ sending $x \in X$ to the equivalence class containing x is called the *quotient function* under R .

We see immediately that the quotient map under R is surjective.

Finally, we consider quotients of topological spaces.

Let (X, τ) be a topological space, and R an equivalence relation on (the underlying set of) X . Let $q : X \rightarrow X/R$ denote the quotient function.

EXERCISE II.4.15. Show that the set

$$\tau_R := \{U \subset X/R : q^{-1}(U) \in \tau\}$$

is a topology on X/R . Furthermore, if τ'_R is a topology on X/R , then $\tau'_R \subset \tau_R$ if and only if the quotient mapping

$$q : (X, \tau) \rightarrow (X/R, \tau'_R)$$

is continuous.

DEFINITION II.4.16. The topological space $(X/R, \tau_R)$ is called the *quotient topological space* under R .

EXAMPLE II.4.17. Let $X = [0, 1]$ with the subspace topology as a subspace of $\mathbb{R}$. The topology of $\mathbb{R}$ is the usual one: the one generated by open intervals.

Define a relation R on X as follows: $(t, s) \in R$ if and only if $t = s$, or $(t, s) = (0, 1)$, or $(t, s) = (1, 0)$.

Then the quotient space X/R is homeomorphic to the circle. Here the circle is regarded as a subset of $\mathbb{C}$ and therefore has the subspace topology.

EXERCISE II.4.18. Prove the statement of Example II.4.17.

DEFINITION II.4.19 (Projective spaces). Let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$. Let “ $\sim$ ” be the relation on $\mathbb{F}^{n+1} - \{0\}$ such that $x \sim y$ if and only if $y = \lambda x$ for some $\lambda \in \mathbb{F} - \{0\}$. The resulting quotient space, denoted by $\mathbb{P}_{\mathbb{F}}^n$, is called the *projective space* of dimension n over $\mathbb{F}$. For a point $x = (x_0, x_1, \dots, x_n) \in \mathbb{F}^{n+1}$, we usually denotes its equivalence class in $\mathbb{P}_{\mathbb{F}}^n$ by $[x_0 : x_1 : \dots : x_n]$.

For $0 \leq i \leq n$, let U_i be the subset of $\mathbb{P}_{\mathbb{F}}^n$ defined by

$$(II.4.5) \quad U_i := \{[x_0 : \dots : x_n] \in \mathbb{P}_{\mathbb{F}}^n : x_i = 1\}$$

EXERCISE II.4.20. (1) Show that U_i is an open set of $\mathbb{P}_{\mathbb{F}}^n$.

(2) Prove $U_i \cong \mathbb{F}^n$.

(3) Prove $\mathbb{P}_{\mathbb{F}}^n = \bigcup_{i=0}^n U_i$.

EXERCISE II.4.21. A *projective line* in $\mathbb{P}_{\mathbb{F}}^2$ is a subset of the form

$$L = \{[x_0, x_1, x_2] \in \mathbb{P}_{\mathbb{F}}^2 \mid \sum_i a_i x_i = 0\}$$

where $a_i \in \mathbb{F}$ are not all 0.

(1) Show that $L \cap U_0$ is a line in the plane U_0 .

(2) Show that any two different projective lines in $\mathbb{P}_{\mathbb{F}}^2$ have exactly one common point.

II.5. Connected and totally disconnected spaces

DEFINITION II.5.1. Let X be a topological space. We say X is *connected* if there is no subspace Y of X such that Y is both open and closed and $Y \neq X$ and $Y \neq \emptyset$. A space is *disconnected* if it is not connected.

Let $x, y \in X$, let $x \sim y$ denote the equivalence relation on X such that $x \sim y$ if and only if there is a connected subspace of X containing both x and y . (Show that this is indeed an equivalence relation!) The equivalence classes are called the *connected components* of X . In other words, A connected component of X is a subspace Y of X , which is connected and is both open and closed in X .

EXERCISE II.5.2. (1) Show that (a, b) is connected.

(2) Find all connected components of $(0, 1) \cup (2, 3)$.

DEFINITION II.5.3. A space X is *totally disconnected* if all subspaces are disconnected, except the singletons (i.e., a subset consisting of one point).

Discrete sets, by definition, are totally disconnected. We proceed to discuss a more interesting example. For a sequence of spaces X_i , i a nonnegative integer, and $x \in \prod_i X_i$, we denote by $x(i)$ the projection of x on X_i .

DEFINITION II.5.4. Let X_i be a finite set with discrete topology, for i a non-negative integer. Let $f_{i+1}: X_{i+1} \rightarrow X_i$ be a mapping. Let

$$(II.5.1) \quad X := \{x \in \prod_i X_i : f_{i+1}(x(i+1)) = x(i)\}$$

be a subspace of $\prod_i X_i$ with the product topology.

A *sequentially profinite set* is a space homeomorphic to the above X for some X_i and f_i .

EXAMPLE II.5.5. Let $X_0 = \{\infty\}$ For a positive integer n , let

$$X_n := \{1, \dots, n, \infty\}$$

and define

$$f_{n+1}: X_{n+1} \rightarrow X_n, f_{n+1}(k) = \begin{cases} k, & k < n+1, \\ \infty, & \text{else.} \end{cases}$$

The resulting sequentially profinite set is denoted by $\mathbb{N}_{\infty}$. The element $n^* \in \mathbb{N}_{\infty}$ is defined by

$$n^*(i) = \begin{cases} n, & i \geq n, \\ \infty, & i < n. \end{cases}$$

The element ∞^* is defined by $\infty^*(i) = \infty$ for all i .

EXERCISE II.5.6. (In this exercise, $\mathbb{N}$ does not include 0.) For any set S , let $\mathcal{P}(S)$ denote the power set of S . Let $W = \mathbb{N} \cup \{\infty\}$, and let $\tau \subset \mathcal{P}(W)$ be defined by

$$\tau = \mathcal{P}(\mathbb{N}) \cup \{U \cup \{\infty, n, n+1, \dots\} \mid U \in \mathcal{P}(\mathbb{N}), n \in \mathbb{N}\}$$

- (1) Show that τ is a topology on W .
- (2) Show that (W, τ) is totally disconnected.
- (3) Show that the mapping

$$\phi: W \rightarrow \mathbb{N}_\infty$$

defined by $\phi(n) = n^*$ and $\phi(\infty) = \infty^*$ is a homeomorphism.

- (4) Let (Y, ρ) be a metric space. Let $\{y_n\}_{n \in \mathbb{N}}$ be a sequence in Y , and let $y \in Y$. We say that the sequence $\{y_n\}$ converges to y and write

$$(II.5.2) \quad \lim_{n \rightarrow \infty} y_n = y$$

if for any $\epsilon > 0$, there is a positive integer $m > 0$, such that for $n > m$, we have $\rho(y_n, y) < \epsilon$. Show that (II.5.2) holds if and only if the mapping

$$\mathbb{N}_\infty \rightarrow Y, \quad n^* \mapsto y_n, \quad \infty^* \mapsto y$$

is continuous.

EXAMPLE II.5.7. Let $n > 1$ be an integer. Let $X_i := \mathbb{Z}/n^i$ and

$$f_i: X_i \rightarrow X_{i-1}, \quad a \pmod{n^i} \mapsto a \pmod{n^{i-1}}.$$

The resulting sequentially profinite set is denoted by $\mathbb{Z}_n$. When n is a prime number, a member in $\mathbb{Z}_n$ is called an n -adic integer.

For $a, b \in \mathbb{Z}_n$, we have $a + b \in \mathbb{Z}_n$ defined by

$$(a + b)(i) := a(i) + b(i)$$

and $a \cdot b$ defined by

$$(a \cdot b)(i) = a(i)b(i).$$

Both the summation and the product are continuous. Indeed, n -adic numbers form a ring which plays an important role in modern number theory.

We may rewrite an element $a \in \mathbb{Z}_n$ as an infinite sum

$$\sum_{k=0}^{\infty} x_k n^k$$

where

$$a(i) \equiv \sum_{k=0}^{i-1} x_k n^k \pmod{n^i}.$$

For instance, we have the multiplicative inverse of $1 - n$, which is $\sum_{k=0}^{\infty} n^k$.

EXERCISE II.5.8. (1) Let p be a prime number. For $m, n \in \mathbb{Z}$, Let $\rho_p(m, n) = p^{-k}$, where k satisfies $p^k \mid (m - n)$ and $p^{k+1} \nmid (m - n)$. Show that ρ_p defines a metric on $\mathbb{Z}$.

- (2) For $m, n \in \mathbb{Z}_p$, let $\rho'_p(m, n) = p^{-k}$, where k satisfies $m(i) = n(i)$ for $i < k$ and $m(k) \neq n(k)$. Show that ρ'_p defines a metric on $\mathbb{Z}_p$, for which the associated topology is the profinite topology.

- (3) For $n \in \mathbb{Z}$, let $r_k(n)$ be the residue of n divided by p^k . Define a mapping $\phi: \mathbb{Z} \rightarrow \mathbb{Z}_p$ by

$$\phi(n) = (r_1(n), r_2(n), \dots).$$

Show that ϕ is well defined and injective.

- (4) Show that for any sequence $\{a_n\}_n$ in $\mathbb{Z}$, the sequence $\{\phi(a_n)\}_n$ in $\mathbb{Z}_p$ converges to a unique element in $\mathbb{Z}_p$. In other words, $\mathbb{Z}_p$ is the *completion* of $\mathbb{Z}$ with respect to the metric ρ_p .

Let X , and X_n be as in Definition II.5.3. For $a \in X_n$, let

$$(II.5.3) \quad U(a) := \{x \in X : x(n) = a\}.$$

EXERCISE II.5.9. Show that the open sets of the form $U(a)$ are a basis for X .

EXERCISE II.5.10. Show that $U(a)$ is open and closed in X .

PROPOSITION II.5.11. *A sequentially profinite set is totally disconnected.*

EXERCISE II.5.12. Prove Proposition II.5.11. (Hint: Use Exercise II.5.10).

II.6. Hausdorff spaces and (quasi) compact spaces

DEFINITION II.6.1. Let X be a topological space, $\{x_n\}_n$ a sequence in X , and let $x \in X$. We say that $\{x_n\}_n$ *converges* to x if, for any open set U such that $x \in U$, there is an integer N such that for all $n > N$, there is $x_n \in U$. We say that $\{x_n\}_n$ is *convergent* if it converges to a point.

EXERCISE II.6.2. Let X be a metric space, $\{x_n\}_n$ a sequence in X , and let $x \in X$. Show that $\lim_{n \rightarrow \infty} x_n = x$ holds if and only if $\{x_n\}_n$ converges to x in the sense of Definition II.6.1.

In a metric space, the limit of a sequence is unique, if exists. However, for a general topological space X , a sequence may converge to more than one points. For instance, if x, y can not be separated by open sets, i.e., an open set contains x if and only if it contains y , then any sequence converging to x also converges to y , and the limit is not necessarily unique. This is one of the reasons that we introduce the following terminology.

DEFINITION II.6.3. A topological space X is *Hausdorff* if for any $x, y \in X$, $x \neq y$ there are open set U, V , satisfying $x \in U$, $y \in V$, and $U \cap V = \emptyset$.

REMARK II.6.4. The property of being Hausdorff is one of the *separation axioms* of topological spaces, conventionally denoted T_i for $0 \leq i \leq 2$. For instance, a topological X space is T_0 , or *Kolmogorov*, if for any two distinct points of X , at least one of them has a neighborhood not containing the other. It is T_1 if for any two distinct points, each has a neighborhood not containing the other. It is T_2 if it is Hausdorff.

EXERCISE II.6.5. Show that a metric space is a Hausdorff topological space.

EXERCISE II.6.6. Show that $\mathbb{R}$ equipped with the Zariski topology is not Hausdorff.

EXERCISE II.6.7 (Line with double origin). Let $X_1 = X_2 = \mathbb{R}$ with the usual topology. For $i = 1, 2$, a point in X_i is denoted by $[a, i]$ where $a \in \mathbb{R}$. Let $X_1 \sqcup X_2$ be equipped with the disjoint union topology.

Let the equivalence relation “ $\sim$ ” on $X_1 \sqcup X_2$ be such that $[a, i] \sim [b, j]$ if and only if $[a, i] = [b, j]$, or $a = b \neq 0$. Let

$$X = X_1 \sqcup X_2 / \sim.$$

Denote the equivalence class of $[0, i]$ by 0_i .

- (1) Show that each $x \in X$ has a neighborhood which is homeomorphic to an open interval.
- (2) Show that X is not Hausdorff.
- (3) Let U be an open set in X , such that $0_0, 0_1 \in U$. Let Y be a Hausdorff space. Let $f: U \rightarrow Y$ be a continuous mapping. Show that $f(0_0) = f(0_1)$.

PROPOSITION II.6.8. A sequence $\{x_n\}_n$ in a Hausdorff space converges to at most one point x , in which case we write $\lim_n x_n = x$. $\square$

EXERCISE II.6.9. Prove Proposition II.6.8.

We proceed to introduce (quasi-)compact topological spaces, which is a natural generalization of compact subsets of Euclidean spaces.

DEFINITION II.6.10. Let X be a topological space. An *open cover* of X is a collection of open sets of X , denoted by $\{U_i\}_{i \in I}$ satisfying $X = \bigcup_{i \in I} U_i$. A *subcover* of an open cover $\{U_i\}_{i \in I}$ of X is a subset of $\{U_i\}_{i \in I}$ which is a cover of X . A *finite (sub)cover* is a (sub)cover that contains only finitely many members.

DEFINITION II.6.11. A topological space X is *quasi-compact* if every open cover of X has a finite subcover. A topological space is *compact* if it is quasi-compact and Hausdorff.

PROPOSITION II.6.12. Any quotient space of a quasi-compact space is quasi-compact. $\square$

EXERCISE II.6.13. Prove Proposition II.6.12.

EXERCISE II.6.14 (*). Let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$. Show that $\mathbb{P}_{\mathbb{F}}^n$ is compact. (Hint: Apply the conclusion of Exercise II.6.13.)

We proceed to consider some concepts related to (quasi-)compact topological spaces.

DEFINITION II.6.15. A topological space is *limit point compact* if any infinite subset A of X has a limit point (Definition II.3.5).

PROPOSITION II.6.16. A quasi-compact space is limit point compact.

PROOF. Suppose X is a quasi-compact space and A is an infinite subset of A . We argue by contradiction. Suppose A has no limit point. For any point in $x \in X$, if $x \notin A$, there is a neighborhood U_x of x such that $U_x \cap A = \emptyset$. For any point $x \in A$, there is a neighborhood U_x of x such that $U_x \cap A = \{x\}$.

The collection $C = \{U_x : x \in X\}$ is an open cover of X since X is quasi-compact, there is a finite subcover of C . However, any subcover contains only finitely many points of A , which is absurd. $\square$

The converse statement of Proposition II.6.16 is false.

EXAMPLE II.6.17 ([Mun00], §28). Let $\mathbb{N}$ be equipped with the discrete topology. Let $Y = \{y_1, y_2\}$ given the trivial topology. Show that $\mathbb{N} \times Y$ is limit point compact but not quasi-compact.

DEFINITION II.6.18. Let X be a topological space. We say that X is *sequentially compact* if every sequence in X has a convergent subsequence.

EXERCISE II.6.19. Show that a limit point compact metric space is sequentially compact.

REMARK II.6.20. The statement of Exercise II.6.19 does not hold without the Hausdorff condition.

PROPOSITION II.6.21. *A metric space is compact if and only if it is sequentially compact.*

The proof of Proposition II.6.21 requires two intermediate results. In mathematical writing, intermediate results that are mainly used to prove a more significant result are called lemmas.

To state the first lemma, we note that the *diameter* of a subset A of a metric space is the supremum of the distance of two points of A .

LEMMA II.6.22. *Let (X, ρ) be a sequentially compact metric space, and let $\mathcal{U} = \{U_i\}_{i \in I}$ is an open cover of X . Then there is a $\delta > 0$ such that for any subset A of X with diameter less than δ , we have $A \subset U_i$ for some i .*

PROOF. We argue by contradiction. Suppose such a δ does not exist. Then for any positive integer n , there is subset A_n of X , of diameter less than n^{-1} , not a subset of any U_i . Let $x_n \in A_n$. Then the sequence $\{x_n\}_n$ has a subsequence converging to some $x \in X$.

Let $\epsilon > 0$ and $i \in I$ satisfy $\mathbf{B}(x, \epsilon) \subset U_i$. Then for some sufficiently large integer n we have $n^{-1} < \epsilon/2$ and $\rho(x_n, x) < \epsilon/2$. For any $a \in A_n$, we have

$$\rho(a, x) \leq \rho(a, x_n) + \rho(x_n, x) \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Therefore, we have $A_n \subset \mathbf{B}(x, \epsilon) \subset U_i$, a contradiction. $\square$

LEMMA II.6.23. *Let (X, ρ) be a sequentially compact metric space, and let $\epsilon > 0$. Then there is a finite cover $\{\mathbf{B}(x_i, \epsilon) : 1 \leq i \leq N\}$.*

PROOF. We argue by contradiction. Suppose for some $\epsilon > 0$, such a cover does not exist. Then, by induction on n , we have a sequence $\{x_n\}_n$, such that x_n is not in $\mathbf{B}(x_i, \epsilon)$ for any $i < n$. Therefore, $\{x_n\}_n$ has no convergent subsequence, a contradiction. $\square$

PROOF OF PROPOSITION II.6.21. By Proposition II.6.16 and Exercise II.6.19, a quasi-compact topological space is sequentially compact.

Now suppose that X is a sequentially compact metric space, and that $\mathcal{U} = \{U_i\}_{i \in I}$ is an open cover of X . We have to show that $\mathcal{U}$ contains a finite cover. By Lemma II.6.22, there is a $\delta > 0$, such that any ball of radius less than $\delta/2$ is a subset of some U_i . Take $\epsilon = \delta/3$. By Lemma II.6.23, the desired result follows. $\square$

PROPOSITION II.6.24. *A sequentially profinite set is compact.*

PROOF. Let X be a sequentially profinite set of the form (II.5.1).

First, we show that X is Hausdorff. If $x, y \in X$, $x \neq y$, then $x(n) \neq y(n)$ for some n . Then we have

$$x_n \in U(x(n)), y_n \in U(y(n)), U(x(n)) \cap U(y(n)) = \emptyset.$$

Therefore, X is Hausdorff.

Next, we show that X is quasi-compact. By Exercise II.5.9, it suffices to show that an open cover

$$\mathcal{U} := \{U(a_j) : j \in J, a_j \in X_i \text{ for some } i\}$$

where J is an index set, has a finite sub-cover.

Let Y_i be the subset of X_i , whose members are all $y \in X_i$ such that $U(y)$ is not a subset of any finite union of members of $\mathcal{U}$. Notice

$$(II.6.1) \quad U(z) \subset U(f_{i+1}(z)).$$

We argue by contradiction. Suppose $\mathcal{U}$ does not have a finite sub-cover. Then $Y_i \neq \emptyset$ for all i . (Why?)

For $y \in Y_i$ and $U(a) \in \mathcal{U}$, since $U(y)$ is the union of all $U(z)$ for $z \in f_{i+1}^{-1}(y)$, there is at least one $U(z)$ that is not a finite union of members of $\mathcal{U}$. In other words, there is at least one $z \in f_{i+1}^{-1}(y)$ that is in Y_{i+1} . Yet in other words, the mapping

$$f_{i+1} : Y_{i+1} \rightarrow Y_i$$

is surjective.

Let $y \in X$ such that $y(i) \in Y_i$. Then $y \in U(a)$ for some $U(a) \in \mathcal{U}$. But then we have $y(j) = a$ for some j , and therefore $U(y(j)) = U(a)$. By the surjectivity of f_k , there is some $k > i, j$ such that $y(k) \in Y_k$. Then $U(y(k)) \subset U(y(j)) = U(a)$, which is a contradiction. $\square$

EXERCISE II.6.25. Prove (II.6.1).

REMARK II.6.26. Topological spaces which are totally disconnected and compact are called *profinite sets*. We have shown that sequentially profinite sets are profinite sets. Indeed, there is an alternative definition of profinite sets, which identify them as certain subspaces of products of finite sets, much like our definition of sequentially profinite sets.

As of the time these notes are written, the on-going project *Condensed Mathematics* [CS24], an effort to reconstruct and develop many of the existing mathematical theories in a unified way, takes the concept of profinite sets as a fundamental role.

We end this chapter with the statement (without proof) of the following

THEOREM II.6.27 (Tychonoff's theorem). *The product of quasi-compact spaces (with the product topology) is quasi-compact.*

EXERCISE II.6.28. ¹ Let J be a set, X a topological space. For $j \in J$, let $X_j = X$. The Cartesian product $\prod_{j \in J} X_j$ with the product topology is denoted by X^J .

- (1) Show that there is a bijection between X^J and the set of all mappings from J to X .
- (2) By (1), we regard elements in X^J as mappings from J to X . Show that a sequence $\{f_n\}_{n \in \mathbb{N}}$ converge to some $f \in X^J$ if and only if for every $j \in J$, the sequence $\{f_n(j)\}_{n \in \mathbb{N}}$ in X converges to $f(j)$.
- (3) Show that if X is Hausdorff, then so is X^J .

¹This example is contributed by Zhuchao Ji.

(4) Let

$$J = \{(j_1, j_2, \dots, j_k, \dots) \mid j_k \in \mathbb{N}, j_1 < j_2 < \dots < j_k \dots\}.$$

Let $X = \{0, 1\}$ with the discrete topology. For $n \in \mathbb{N}$, define $f \in X^J$ by

$$f_n((j_1, j_2, \dots, j_k, \dots)) = \begin{cases} 1, & \exists k \in \mathbb{N}, j_{2k} = n, \\ 0, & \text{else.} \end{cases}$$

Consider the strictly increasing sequence of positive integers

$$j = (n_1, n_2, \dots, n_k, \dots) \in J.$$

Show that the sequence $\{f_{n_k}\}_{k \in \mathbb{N}}$ in X does not converge.

Hint: Consider the sequences

$$f_{n_1}(j), f_{n_3}(j), f_{n_5}(j), \dots$$

and

$$f_{n_2}(j), f_{n_4}(j), f_{n_6}(j), \dots.$$

Therefore, we conclude that $\{f_n\}_{n \in \mathbb{N}}$ does not have a convergent subsequence. So X^J is not sequentially compact. But by Theorem II.6.27, X^J is quasi-compact. By (3), X^J is compact.

We conclude that compact spaces are not necessarily sequentially compact. In particular, limit point compact spaces are not necessarily sequentially compact.

Much of the material of this section is taken from §26 of [Mun00].

CHAPTER III

Homotopy and fundamental groups

III.1. Homotopy

Recall that a homeomorphism f from one topological space X to another one Y is a bijective continuous function (i.e. a bijective map) with a continuous inverse function. In this case we say that X is homeomorphic to Y and vice versa. We regard X and Y as identical, if we do not consider structures on X and Y other than their topologies. For instance, a disc and a rectangle are homeomorphic.

Sometimes, the concept of homeomorphism is unnecessarily rigid. For instance, the complex plane without the origin, $\mathbb{C}^\times$, is not homeomorphic to the unit circle S^1 centered at 0. However, they have much in common, for instance, when considering the existence problem of a continuous square root function, some trial and error may convince you that the problem “feels the same” for $\mathbb{C}^\times$ and S^1 . This observation leads us to the following concepts.

Throughout the rest of this chapter, I denotes the closed interval $[0, 1]$.

DEFINITION III.1.1. Let X, Y be topological spaces and $f, g : X \rightarrow Y$ be maps. If there is a map

$$F : X \times I \rightarrow Y$$

such that $F(-, 0) = f$ and $F(-, 1) = g$, then F is a *homotopy from f to g* , and f and g are *homotopic (through F)*, written as $f \equiv_F g$ or $f \equiv g$ if F is unspecified.

Given $x_0 \in X$ and $y_0 \in Y$, (which we call *base points*) then F is a *homotopy from f to g relative to x_0* , and also f and g are *homotopic relative to x_0* if, in addition to the above, we have

$$f(x_0) = g(x_0) = y_0, \quad F(x_0, t) = y_0, \quad \forall t \in I,$$

In this case we also write

$$f \equiv_F g : (X, x_0) \rightarrow (Y, y_0).$$

EXERCISE III.1.2. Show that homotopy is an equivalence relation between maps from X to Y .

Show that homotopy relative to x_0 is an equivalence relation between maps from (X, x_0) to (Y, y_0) .

DEFINITION III.1.3. Two spaces X and Y are *homotopy equivalent*, if there are maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $gf \equiv \mathbb{I}_X$ and $fg \equiv \mathbb{I}_Y$.

For $x_0 \in X$ and $y_0 \in Y$, we say X and Y are *homotopy equivalent relative to the base points* if the aforementioned homotopies are relative to base points.

In the aforementioned cases, f and g are called *homotopy equivalences (relative to a base point)*.

A space is *contractible* if it is homotopy equivalent to a singleton; it is *strongly contractible* if the homotopy is relative to the base point.

EXERCISE III.1.4. Show that $\mathbb{R}^n$ and the close disks and open discs in $\mathbb{R}^n$ are strongly contractible.

EXERCISE III.1.5. Show that $\mathbb{C}^\times$ and S^1 are homotopy equivalent relative to a base point, but not homeomorphic to each other.

(Hint: To show the latter, prove that S^1 is compact whereas $\mathbb{C}^\times$ is not.)

We have a more general concept of relative homotopy.

Let notations be as above, we write

$$f : (X, A) \rightarrow (Y, B)$$

to indicate that f is a map from X to Y satisfying $f(A) \subset B$.

DEFINITION III.1.6. Let X be a space with a subspace A , and Y a space with a subspace B . A *relative homotopy* is a map

$$F : X \times I \rightarrow Y$$

such that $F(A \times I) \subset B$. We say $f, g : X \rightarrow Y$ are *homotopic relative to A* , B if $f = F(-, 0)$, $g = F(-, 1)$ for such an F , or simply $f, g : (X, A) \rightarrow (Y, B)$ are homotopic.

EXERCISE III.1.7. Let $f_0, f_1 : (X, A) \rightarrow (Y, B)$, $g_0, g_1 : (Y, B) \rightarrow (Z, C)$ be homotopic. Show that $g_0 f_0, g_1 f_1 : (X, A) \rightarrow (Z, C)$ are relatively homotopic.

III.2. Fundamental groups

This section requires basic terminologies of group theory.

Let $\partial I = \{0, 1\}$.

DEFINITION III.2.1. Let (X, x_0) be a space with base point. Let $\pi_1(X, x_0)$ be the set of relative homotopy classes of maps

$$\sigma : (I, \partial I) \rightarrow (X, x_0).$$

We also call such a map a *loop in X based at x_0* .

We write $\sigma \simeq \tau$ to indicate that σ and τ are loops homotopic relative to ∂I and a base point.

Two loops based at x_0 , $f, g : (I, \partial I) \rightarrow (X, x_0)$, may be concatenated to produce another loop based at x_0 :

$$(III.2.1) \quad \begin{aligned} & \sigma * \tau : (I, \partial I) \rightarrow (X, x_0), \\ & t \mapsto \begin{cases} \sigma(2t), & 0 \leq t \leq 1/2, \\ \tau(2t - 1), & 1/2 \leq t \leq 1. \end{cases} \end{aligned}$$

EXERCISE III.2.2. Show that if $\sigma \simeq \sigma'$, $\tau \simeq \tau'$, then $\sigma * \sigma' \simeq \tau * \tau'$.

DEFINITION III.2.3. Let $\alpha, \beta \in \pi_1(X, x_0)$. Let $\sigma \in \alpha$ and $\tau \in \beta$. Then the *composition* of α and β is defined as

$$\alpha\beta := [\sigma * \tau].$$

REMARK III.2.4. By Exercise III.2.2, the composition is independent of the choices of σ and τ .

Let $c_{x_0} \in \pi_1(X, x_0)$ be the class containing constant loop based at x_0 .

EXERCISE III.2.5. Let $\alpha \in \pi_1(X, x_0)$. Show that

$$\alpha c_{x_0} = c_{x_0} \alpha = \alpha.$$

EXERCISE III.2.6. Let $\alpha, \beta, \gamma \in \pi_1(X, x_0)$. Show that

$$(\alpha\beta)\gamma = \alpha(\beta\gamma).$$

EXERCISE III.2.7. Let $[\sigma] = \alpha \in \pi_1(X, x_0)$. Consider the path

$$\sigma' : (I, \partial) \rightarrow (X, x_0), \quad t \mapsto \sigma(1-t)$$

and let $\alpha^{-1} = [\sigma']$. Show that $\alpha\alpha^{-1} = c_{x_0}$.

We have shown that the set $\pi_1(X, x_0)$ form a group with respect to the composition. We will write 1 for c_{x_0} when no ambiguity is likely.

DEFINITION III.2.8. The group with underlying set $\pi_1(X, x_0)$ and the composition discussed above is called the *fundamental group* of X with base point x_0 .

We proceed to study the significance of the end point.

EXERCISE III.2.9. Let X be a space. For $x, y \in X$, we say that x and y are connected by a path σ if there is a map

$$\sigma : I \rightarrow X$$

satisfying $\sigma(0) = x$ and $\sigma(1) = y$. If this is the case, we write $x \simeq y$. Show that “ $\simeq$ ” is an equivalence relation.

DEFINITION III.2.10. A *path component* of a space X is an equivalence class with respect to the relation “ $\simeq$ ” as in Exercise III.2.9.

The space X is *path-connected* if it has only one path component.

EXERCISE III.2.11. Show that a path-connected space is connected. (Hint: Show that I is connected.)

EXAMPLE III.2.12. The *Warsaw sine curve*

$$T = \{(x, \sin x) : x \in (0, 1)\} \cup \{(0, 0)\}$$

is connected but not path-connected.

Suppose we have $x_0, x_1 \in X$ and a path σ in X such that $\sigma(x_i) = x_i$ for $i = 0, 1$. Then we have a function of sets

$$(III.2.2) \quad \begin{aligned} \sigma_{\#} : \pi_1(X, x_1) &\rightarrow \pi_1(X, x_0) \\ [\tau] &\mapsto [\sigma\tau\sigma']. \end{aligned}$$

EXERCISE III.2.13. Show that the function (III.2.2) is well defined and is an isomorphism of groups.

Therefore, we have proved the following

PROPOSITION III.2.14. *Let X be a path connected space. Up to group isomorphism, $\pi_1(X, x_0)$ is independent of the choice of x_0 .* $\square$

Let $f : X \rightarrow Y$ be a map.

DEFINITION III.2.15. The group

$$\pi_1(f) : \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0)), \quad [\sigma] \mapsto [f\sigma]$$

is called the *homomorphism induced by f* . Alternatively, we denote $\pi_1(f)$ by f_* .

EXERCISE III.2.16. Show that f_* is a well-defined mapping and a homomorphism of groups.

PROPOSITION III.2.17. *Let $f, g: (X, x_0) \rightarrow (Y, y_0)$ be maps homotopic relative to the base point. Then $f_* = g_*$.*

PROOF. The proof is formal. $\square$

PROPOSITION III.2.18. (1) *For a space X and a base point x_0 , we have*

$$\mathbb{I}_{\pi_1(X, x_0)} = (\mathbb{I}_X)_*.$$

(2) *For maps $f: X \rightarrow Y$ and $g: Y \rightarrow Z$, we have $(gf)_* = g_*f_*$.*

PROOF. The proof is formal. $\square$

REMARK III.2.19. By Proposition III.2.17 and Exercise III.1.7, it makes sense to write α_* and $\alpha_*\beta_*$ as the induced homomorphisms, where α, β_* are homotopy classes of maps relative to base points.

In the terminology of category theory, Proposition III.2.18 says that taking fundamental group is a “functor” from the “(homotopy) category of spaces” to “the category of groups.”

COROLLARY III.2.20. *If $f: X \rightarrow Y$ is a homotopy equivalence relative to base points, then f_* is an isomorphism.*

COROLLARY III.2.21. *If X is strongly contractible, then $\pi_1(X, x_0) = 0$.*

A path-connected space with trivial fundamental group is said to be *simply connected*. There are many examples of simply connected and not contractible spaces. For instance, the sphere S^n with $n > 1$ is such a space. However, the proof of this fact is beyond the scope of this course.

The fact $\pi_1(S^n, x) = 0$ for $n > 1$ is visually clear, as one can easily imagine a loop on S^n , or more concretely, on S^2 to collapse to the base point without leaving the sphere. A rigorous proof relies on the following

LEMMA III.2.22. *A path*

$$\sigma: (I, \partial I) \rightarrow (S^n, x)$$

is (relatively) homotopic to a path σ' which is not surjective.

Lemma III.2.22 is an immediate corollary of the simplicial approximation theorem.

EXERCISE III.2.23. Assuming Lemma III.2.22, show $\pi_1(S^2, x) = 0$. (Hint: First show that the S^2 minus a point is homeomorphic, via the Stereographic projection, to $\mathbb{R}^2$, which is contractible.)

PROPOSITION III.2.24. *Let $p_i: X_1 \times X_2 \rightarrow X_i$, $i = 1, 2$ be the projections. Then*

$$((p_1)_*, (p_2)_*): \pi_1(X_1 \times X_2, (x_1, x_2)) \rightarrow \pi_1(X_1, x_1) \times \pi_1(X_2, x_2)$$

is an isomorphism.

PROOF. The proof is formal. $\square$

EXERCISE III.2.25. Let $H: X \times I \rightarrow Y$ be a homotopy (not necessarily relative to base points) from f_0 to f_1 . Let $x_0 \in X$, and let $y_i = f_i(x_0)$ for $i = 0, 1$. Let $h: I \rightarrow Y$ be the path defined by $h(t) = H(x_0, t)$. Consider the “change of base point” homomorphism

$$h_{\#}: \pi_1(Y, y_1) \rightarrow \pi_1(Y, y_0).$$

as defined in (III.2.2). Show that

$$f_{0*} = h_{\#} f_{1*}: \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0).$$

COROLLARY III.2.26. *Let $f: X \rightarrow Y$ is a homotopy equivalence (not necessarily relative to base points). Let $x_0 \in X$, and let $y_0 = f(x_0)$. Then*

$$f_*: \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$$

is an isomorphism.

COROLLARY III.2.27. *If X is contractible, then $\pi_1(X, x_0) = 0$ for any $x_0 \in X$. (Notice that the contraction in this case is not necessarily relative to base points.)*

EXERCISE III.2.28. Prove Corollary III.2.26 and Corollary III.2.27 using the conclusion of Exercise III.2.25.

III.3. The fundamental group of the circle

We present a concrete example of the (nontrivial) fundamental group of a space which is well understood.

Let S^1 be the unit circle in the complex plane with base point 1. The set $\{0, 1\}$ is denoted by ∂I . For each $n \in \mathbb{Z}$, define the loop

$$\sigma_n: (I, \partial I) \rightarrow (S^1, 1), \quad t \mapsto \exp(2\pi i n t).$$

The primary objective of this chapter is to prove the following

THEOREM III.3.1. *The function*

$$\Phi: \mathbb{Z} \rightarrow \pi_1(S^1, 1), \quad n \mapsto [\sigma_n]$$

is an isomorphism of groups.

DEFINITION III.3.2. For a map $f: S^1 \rightarrow S^1$, the *degree* of f is the integer $\Phi^{-1}([f])$.

Intuitively, the function Φ^{-1} counts how many times the interval I gets wrapped around the circle (with an orientation) by a loop σ , and the class $[\sigma]$ is completely determined by this number. The critical challenge of this section is to mathematically formalize this intuitive idea.

Let $p: \mathbb{R} \rightarrow S^1$ be the map defined by $p(t) = \exp 2\pi i t$. Then p preserves base points, whenever the base point of $\mathbb{R}$ is an integer.

The following lemma is the heart of the proof of Theorem III.3.1.

LEMMA III.3.3. *Let K be a connected quasi-compact topological space. For any map*

$$f: K \times I \rightarrow S^1$$

and

$$f': K \rightarrow \mathbb{R}$$

such that $p f'(x) = f(x, 0)$ for all $x \in K$, there is a unique map

$$\tilde{f}: K \times I \rightarrow \mathbb{R}$$

satisfying $f'(-) = \tilde{f}(-, 0)$ and $f = p\tilde{f}$.

REMARK III.3.4. The lemma holds even without the connectedness or the quasi-compactness assumption on K .

PROOF. For $x \in S^1$, let $U_x := S^1 - \{-x\}$. Then U_x is a neighborhood of x . Given $y \in p^{-1}(-x) \cap [0, 1)$, we have

$$p^{-1}(U_x) = \mathbb{R} - \{y + n : n \in \mathbb{Z}\}.$$

Let $V_x^j := (y + j, y + j + 1)$. Then the restriction of p on V_x^j is an isomorphism from V_j to U_x . Denote its inverse by

$$q_x^j : U_x \rightarrow V_x^j.$$

Case 1: $K = \{z\}$. It suffice to show that, for any map

$$f : (I, \partial I) \rightarrow (S^1, 1)$$

and a fixed $n \in \mathbb{Z}$, there is a unique map

$$\tilde{f} : I \rightarrow \mathbb{R}$$

satisfying $p\tilde{f} = f$ and $\tilde{f}(0) = n$.

Suppose $[0, \epsilon) \subset f^{-1}(U_1)$. Then we have $n \in V_1^j$ for some $j \in \mathbb{Z}$. For $t \in [0, \epsilon)$, let

$$\tilde{f}(t) = q_1^j f(t).$$

Then we have defined

$$(III.3.1) \quad \tilde{f} : [0, \epsilon) \rightarrow X.$$

Let L denote the set of all $l \in (0, 1)$ such that there is a map

$$\tilde{f} : [0, l) \rightarrow \mathbb{R}$$

satisfying $p\tilde{f} = f$ and $\tilde{f}(x, 0) = f'(x)$. By (III.3.1), $L \neq \emptyset$.

Suppose the supremum of L is $c < 1$. Let $[a, b)$ be a small interval such that $c \in [a, b)$ and $f^{-1}[a, b) \subset U_{f(c)}$.

By the definition of c , since $a < c$, $\tilde{f}$ is defined on $[0, a]$. Therefore, we have $j \in \mathbb{Z}$ such that $\tilde{f}(a) \in V_{f(a)}^j$. For $t \in [a, b)$, defined

$$\tilde{f}(t) = q_{f(a)}^j f(t).$$

Then we have defined $\tilde{f}(t)$ for $t \in [0, b)$, which is a contradiction. We have thus proved $c = 1$, and the fact that $\tilde{f}(t)$ is defined for $t = c$. Therefore, the existence of $\tilde{f}$ is justified.

For the uniqueness, notice that for $\tilde{f}_1$ and $\tilde{f}_2$ satisfying

$$p\tilde{f}_1 = p\tilde{f}_2 = f, \quad \tilde{f}_1(0) = \tilde{f}_2(0),$$

we have $\tilde{f}_1(t) - \tilde{f}_2(t) = n$ where $n \in \mathbb{Z}$. Taking $t = 0$, we have $n = 0$.

Case 2: the general case. For $z \in K$, $t \in I$, let $C_{z,t} = A_{z,t} \times B_{z,t}$ where $A_{z,t}$ is a neighborhood of z and $B_{z,t}$ is a small open neighborhood of t in I such that

$$C_{x,t} \subset f^{-1}(U_{f(z,t)}).$$

Fix $t \in I$, $t \neq 1$. Since K is compact, there is a finite cover of $K \times \{t\}$, $\{C_{z_i,t} \mid 1 \leq i \leq m\}$. Let

$$W_t = \cap_{i=1}^m B_{z_i}.$$

Then W_t is an open interval (unless $t = 0$ or 1 , in which case it is half open).

For $1 \leq i \leq m$, there is a $j \in \mathbb{Z}$, such that $f'(z_i) \in V_1^j$. For $(z, t) \in C_{z_i, 0}$, let

$$\tilde{f}_i(z, t) = q_1^j f(z, t).$$

Then we have $p\tilde{f}_i(z, t) = f(z, t)$ and $\tilde{f}_i(z, 0) = f'(z)$. By the uniqueness in Case 1, we have

$$\tilde{f}_{i_1}(z, t) = \tilde{f}_{i_2}(z, t), \quad (z, t) \in C_{z_{i_1}} \cap C_{z_{i_2}}.$$

Therefore, we define $\tilde{f}(z, t)$ for $t \in W_0$ by

$$(III.3.2) \quad \tilde{f}(z, t) = \tilde{f}_i(z, t), \quad (z, t) \in C_{z_i, 0}.$$

As in Case 1, let L denote the set of all $l \in (0, 1)$ such that there is a map

$$\tilde{f}: K \times [0, l) \rightarrow \mathbb{R}$$

satisfying $p\tilde{f} = f$ and $\tilde{f}(x, 0) = f'(x)$. By (III.3.2), $L \neq \emptyset$. The rest of the proof follows exactly as in Case 1. $\square$

Let K be a singleton or $K = I$, we have the following corollaries.

COROLLARY III.3.5. *Let*

$$\sigma: (I, \partial I) \rightarrow (S^1, 1)$$

be a loop. Then there is a unique path

$$\tilde{\sigma}: (I, 0) \rightarrow (\mathbb{R}, 0)$$

satisfying $p\tilde{f} = f$.

COROLLARY III.3.6. *Let*

$$\sigma_0, \sigma_1: (I, \partial I) \rightarrow (S^1, 1)$$

be loops, and let F be a homotopy from σ_0 to σ_1 relative to the base points. Then there is a homotopy $\tilde{F}$ from $\tilde{\sigma}_0$ and $\tilde{\sigma}_1$ such that $p\tilde{F} = F$ and $\tilde{F}(0, s) = 0$ for all $s \in I$.

LEMMA III.3.7. *Consider the loops*

$$\sigma_0, \sigma_1: (I, \partial I) \rightarrow (S^1, 1),$$

and their lifts $\tilde{\sigma}_i$, such that $p\tilde{\sigma}_i = \sigma_i$ and $\tilde{\sigma}_i(0) = 0$ for $i = 0, 1$. Then σ_0 and σ_1 are homotopic relative to the end points if and only if $\tilde{\sigma}_0(1) = \tilde{\sigma}_1(1)$.

PROOF. Suppose we have a homotopy

$$F: I \times I \rightarrow S^1$$

such that $F(t, i) = \sigma_i(t)$, $i = 0, 1$, and $F(0, s) = F(1, s) = 1$ for all $s \in I$. We have to show $\tilde{\sigma}_0(1) = \tilde{\sigma}_1(1)$.

Indeed, by Corollary III.3.6, we have

$$\tilde{F}: I \times I \rightarrow \mathbb{R}$$

such that $\tilde{F}(-, \epsilon) = \tilde{\sigma}_\epsilon$, $\epsilon = 0, 1$, and $\tilde{F}(1, s) \in \mathbb{Z}$. Therefore, $\tilde{F}(1, -)$ is constant and we have

$$\tilde{\sigma}_0(1) = \tilde{F}(1, 0) = \tilde{F}(1, 1) = \tilde{\sigma}_1(1).$$

Conversely, suppose $\tilde{\sigma}_0(1) = \tilde{\sigma}_1(1)$. Then

$$F(t, s) := p((1-s)\tilde{\sigma}_0(t) + s\tilde{\sigma}_1(t))$$

is the required homotopy. $\square$

Since $p\tilde{\sigma}(1) = 1$, we have $\tilde{\sigma}(1) \in \mathbb{Z}$. Therefore, Lemma III.3.7 yields a function

$$\Psi: \pi_1(S^1, 1) \rightarrow \mathbb{Z}, [\sigma] \mapsto \tilde{\sigma}(1).$$

The proof of Theorem III.3.1 is concluded by the following

EXERCISE III.3.8. The function Ψ is the inverse of Φ .

REMARK III.3.9. We will omit the base points in the notation of fundamental groups whenever no risk of confusion is likely.

It is not hard to verify that the torus T is homeomorphic to $S^1 \times S^1$. By Proposition III.2.24, we have

$$\pi_1(T) \cong \mathbb{Z} \times \mathbb{Z}.$$

On the other hand, by Exercise III.2.23, we have $\pi_1(S^2) = 0$. Therefore, T and S are not homotopy equivalent, and in particular, not homeomorphic.

REMARK III.3.10. We revisit the notions of powers of uni-modular complex numbers and winding numbers, as discussed in Chapter I.

As revealed in this section, the path

$$I \rightarrow S^1: t \mapsto \exp(2\pi it)^n = \exp(2\pi it)$$

represents the integer n in $\pi_1(S^1, 1)$.

Consider a polyline L and a point P in $\mathbb{R}^2$ such that $P \notin L$. Exercise III.1.5 shows $\mathbb{R}^2 - P \simeq S^1$, and the winding number of L around P is

$$[\sigma] \in \pi_1(\mathbb{R}^2 - P) \cong \mathbb{Z}$$

where σ is a path that travels along L .

In the first case, to obtain n , we refer to the arithmetic of complex numbers. In the second case, to obtain the winding number, we refer to the linear structure of L as well as $\mathbb{R}^2$. In this section, we unified these two quantities as classes in $\pi_1(S^1)$. It is remarkable that, in doing so, we work only with topological spaces and continuous maps, which is much less structure than the arithmetic of complex number or the linear structure of $\mathbb{R}^2$.

III.4. The n th roots of complex numbers and the logarithmic function

In this section we discuss the logarithmic and the roots of unity in $\mathbb{C}$, the latter of which is discussed in Chapter I.

Fix an integer $n > 0$, let the n th power function be denoted by $p_n: S^1 \rightarrow S^1$.

PROPOSITION III.4.1. *For $n \neq \pm 1$, the map p_n has no continuous right or left inverse. In other words, there is no continuous n th root of unity map from S^1 to itself.*

PROOF. If the continuous right inverse, say ξ_n exists, passing to fundamental groups, this says that we have

$$\mathbb{I}_{\pi_1(S,1)} = (\xi_n p_n)_* = (\xi_n)_* (p_n)_*: \pi_1(S, 1) \rightarrow \pi_1(S, 1).$$

Recall the isomorphism of groups $\Phi: \mathbb{Z} \rightarrow \pi_1(S^1, 1)$ and $\Psi: \pi_1(S^1, 1) \rightarrow \mathbb{Z}$. A routine computation shows that the homomorphism

$$\Psi(p_n)_* \Phi: \mathbb{Z} \rightarrow \mathbb{Z}$$

is the multiplication by n . Therefore, the homomorphism

$$\Psi(\xi_n)_* \Phi: \mathbb{Z} \rightarrow \mathbb{Z}$$

is the inverse of $\Psi p_n \Phi$. Therefore, $\Psi(\xi_n)_* \Phi$ is a homomorphism taking n to 1, which does not exist. This yields a contradiction.

The non-existence of a continuous left inverse can be proved similarly. $\square$

The function $z \mapsto \exp(2\pi iz)$ is a map from $\mathbb{R}$ to S^1 .

PROPOSITION III.4.2. *The map $z \mapsto \exp(2\pi iz)$ does not have a continuous right inverse. (A right inverse of a map $f: X \rightarrow Y$ is a function $g: Y \rightarrow X$ such that fg is the identity of map Y . Notice that a right inverse is not necessarily an inverse.)*

In other words, there is no continuous logarithmic from S^1 to $\mathbb{R}$.

EXERCISE III.4.3. Prove Proposition III.4.2.

III.5. The fundamental theorem of algebra

We present the proof of the fundamental theorem of algebra by Hatcher [Hat02] based on the fundamental group of S^1 .

THEOREM III.5.1 (The Fundamental Theorem of Algebra). *Every non-constant polynomial over $\mathbb{C}$ has a root.*

PROOF. Let $p(z) = z^n + \sum_{k=1}^n a_k z^{n-k}$ be a non-constant polynomial. Suppose, for a contradiction, that $p(z)$ has no root. Then for $r \geq 0$, we define

$$f_r: (I, \partial I) \rightarrow (S^1, 1), \quad f_r(s) = \frac{p(re^{2\pi is})/p(r)}{|p(re^{2\pi is})/p(r)|}.$$

Then we have $f_r \simeq f_0$ relative to ∂I . Since $f_0 = c_1$ is the constant path, we have

$$(III.5.1) \quad [f_r] = [f_0] = 1 \in \pi_1(S^1, 1).$$

Fix an r such that

$$r > \max\{|a_1| + \cdots + |a_n|, 1\}.$$

For $z \in \mathbb{C}$, $|z| = r$, we have

$$|z|^n = r^n = r \cdot r^{n-1} > (|a_1| + \cdots + |a_n|)r^{n-1} > \left| \sum_{k=1}^n a_k z^{n-k} \right|.$$

Therefore, for $t \in I$, the polynomial

$$p_t(z) = z^n + t \sum_{k=1}^n a_k z^{n-k}$$

has no root of length r . Therefore, in the definition of f_r , we may replace $p(z)$ with $p_t(z)$ and obtain a homotopy (relative to ∂I) from f_r to the map

$$(I, \partial I) \rightarrow (S^1, 1), \quad s \mapsto \frac{p_0(re^{2\pi is})/p_0(r)}{|p_0(re^{2\pi is})/p_0(r)|} = e^{2\pi i n s}.$$

Therefore, we have

$$[f_r] = [\sigma_n] = \Phi(n) \in \pi_1(S^1, 1),$$

a contradiction to (III.5.1). $\square$

CHAPTER IV

Projective Geometry

IV.1. The projective line and the projective plane

Classically, the projective space $\mathbb{P}_{\mathbb{F}}^n$ are regarded as $\mathbb{F}^n$ adjoining “a point of infinity” for each line through the origin. The idea is to identify $\mathbb{F}^n$ with one of the open sets U_i of $\mathbb{P}_{\mathbb{F}}^n$, say U_0 . Then the complement

$$\mathbb{P}_{\mathbb{F}}^n - U_0 = \{[0: x_1: \cdots: x_n]\}$$

is regarded as the “points at infinity”.

For $n = 1$, there is only one “point at infinity”.

EXAMPLE IV.1.1 (the one-point compactification of $\mathbb{R}^n$). Consider the set $X = \mathbb{R}^n \cup \{*\}$. Define a topology τ_X on X by

$$(IV.1.1) \quad \tau_X := \tau_{\mathbb{R}^n} \cup \{U \cup \{*\}: \mathbb{R}^n - U \text{ is compact.}\}$$

The space (X, τ_X) is called the *one-point compactification* of $\mathbb{R}^n$

EXERCISE IV.1.2. (1) Show that (IV.1.1) indeed defines a topology on X , and show that X is homeomorphic to S^n , the n -dimensional unit sphere.

(2) Prove $\mathbb{P}_{\mathbb{R}}^1 \cong S^1$ and $\mathbb{P}_{\mathbb{C}}^1 \cong S^2$.

Consider the case $n = 2$. You may have heard the saying “Two parallel lines on a plane intersect at infinity”. In the space $\mathbb{P}_{\mathbb{R}}^2$ this can be made into a mathematically rigorous statement. First, let us make clear what is a line in $\mathbb{P}_{\mathbb{R}}^2$. Let

$$(IV.1.2) \quad \begin{aligned} L' &:= \{(x_1, x_2) \in \mathbb{R}^2: \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 = 0\} \\ &:= \{[1: x_1: x_2] \in \mathbb{P}_{\mathbb{R}}^2: \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 = 0\}, \quad (\alpha_1, \alpha_2) \neq (0, 0). \end{aligned}$$

be a line in $\mathbb{R}^2$, which is identified to U_0 . Let $\xi = (\xi_1, \xi_2) \in L'$, and we have a homeomorphism

$$\mathbb{R} \rightarrow L', \quad t \mapsto (\alpha_2 t, -\alpha_1 t) + \xi = [1: \alpha_2 t + \xi_1: -\alpha_1 t + \xi_2].$$

Since $\mathbb{P}_{\mathbb{R}}^2$ is Hausdorff, we may discuss limits as we do for metric spaces. Therefore we have

$$(IV.1.3) \quad \begin{aligned} &\lim_{t \rightarrow \infty} [1: \alpha_2 t + \xi_1: -\alpha_1 t + \xi_2] \\ &= \lim_{t \rightarrow \infty} [t^{-1}: \alpha_2 + \xi_1 t^{-1}: -\alpha_1 + \xi_2 t^{-1}] \\ &= [0: \alpha_2: -\alpha_1] \end{aligned}$$

which are to be regarded as “points on L' at infinity”.

REMARK IV.1.3. Which of the points in $\mathbb{P}_{\mathbb{F}}^2$ are “at infinity” depends on how you place an Euclidean plane $\mathbb{F}^2$ in $\mathbb{P}_{\mathbb{F}}^2$. For instance, if U_1 is chosen, then the “points at infinity” are of the form $[x_0: 0: x_2]$. The same goes for $\mathbb{P}_{\mathbb{F}}^n$.

EXERCISE IV.1.4. Show that the closure of L in $\mathbb{P}_{\mathbb{R}}^2$ is the set

$$\{[x_0 : x_1 : x_2] \in \mathbb{P}_{\mathbb{R}}^2 : \alpha_0 x_1 + \alpha_1 x_1 + \alpha_2 x_2 = 0\}.$$

The same computation applies to the case of the complex projective plane. We have justified the following definition.

DEFINITION IV.1.5. A *line* in the projective plane $\mathbb{P}_{\mathbb{F}}^2$ is the subspace defined by

$$L = \{[x_0 : x_1 : x_2] \in \mathbb{P}_{\mathbb{F}}^2 : \alpha_0 x_1 + \alpha_1 x_1 + \alpha_2 x_2 = 0\}$$

for some $\alpha_i \in \mathbb{F}$, $i = 1, 2, 3$, not all zero.

REMARK IV.1.6. Notice that many different choices of the numbers x_0, x_1, x_2 represent the same point $[x_0 : x_1 : x_2] \in \mathbb{P}_{\mathbb{F}}^2$. However, whether the equation

$$\alpha_0 x_1 + \alpha_1 x_1 + \alpha_2 x_2 = 0$$

is satisfied is independent of the choices.

REMARK IV.1.7. Notice that the equation for a projective line in $\mathbb{P}_{\mathbb{F}}^2$ looks exactly like the equation of 2-dimensional linear subspace of $\mathbb{F}^3$. We call it the linear subspaces associated to the projective line.

Therefore, we have linearly independent vectors $\xi = (\xi_0, \xi_1, \xi_2)$, $\eta = (\eta_1, \eta_2, \eta_3)$ such that we have the parametrization

$$(IV.1.4) \quad L = \{[s\xi_0 + t\eta_0 : s\xi_1 + t\eta_1 : s\xi_2 + t\eta_2] : s, t \in \mathbb{F}\}.$$

EXERCISE IV.1.8. Show that a projective line in $\mathbb{P}_{\mathbb{F}}^2$ is homeomorphism to $\mathbb{P}_{\mathbb{F}}^1$.

EXERCISE IV.1.9. (1) Show that two distinct lines in $\mathbb{P}_{\mathbb{F}}^2$ intersects at exactly one point.

(2) Show that there is a unique line passing through two given points in $\mathbb{P}_{\mathbb{F}}^2$.

We say that several points on $\mathbb{P}_{\mathbb{F}}^2$ are *collinear* if they lie on a same line.

THEOREM IV.1.10 (Desargues' Theorem). Let $A_i, B_i, C \in \mathbb{P}_{\mathbb{F}}^2$, $i = 1, 2, 3$, such that A_i, B_i, C are collinear and distinct. Let L_{ij} (resp. M_{ij}) be the line connecting A_i and A_j (resp. B_i and B_j). Let P_{ij} be the intersection point of L_{ij} and M_{ij} .

Then, P_{12}, P_{13}, P_{23} are collinear.

PROOF. Throughout this proof, we regard a point in $\mathbb{P}_{\mathbb{F}}^2$ as a 1-dimensional subspace of $\mathbb{R}^3$. Suppose that $\gamma \in \mathbb{R}^3$ generates C , and $\alpha'_i \in \mathbb{R}^3$ generates A_i . Then we have $\lambda^i \in \mathbb{R}$, not all zero for $i = 0, 1, 2$, such that $\beta_i = \lambda^i \alpha'_i + \gamma$ generates B_i .

Let $\alpha_i = \lambda^i \alpha'_i$. Then α_i generates A_i , and we have $\beta_i = \alpha_i + \gamma$.

It follows that P_{ij} is a subspace of the intersection of $\mathbb{R}\{\alpha_i, \alpha_j\}$ and $\mathbb{R}\{\alpha_i + \gamma, \alpha_j + \gamma\}$. A straightforward computation then shows that

$$P_{ij} = \mathbb{R}\{\alpha_i - \alpha_j\}.$$

Therefore, P_{12}, P_{13}, P_{23} is in the same 2 dimensional subspace of $\mathbb{R}^3$, and the proof is concluded. $\square$

REMARK IV.1.11. A high-school-level statement of Desargues' Theorem is more complicated because the cases that L_{ij} and M_{ij} are parallel has to be excluded, and a high-school-level proof requires hard work. In the setting of projective plane, however, both the statement and the proof are remarkably simple.

We close this section with a concept on $\mathbb{P}_{\mathbb{F}}^n$ for any integer $n > 0$.

DEFINITION IV.1.12. Let A be an invertible $n \times n$ matrix. Then A defines a function

$$\phi_A : \mathbb{P}_{\mathbb{F}}^n \rightarrow \mathbb{P}_{\mathbb{F}}^n, [x] \mapsto [(Ax^t)^t],$$

which is called the *projective transformation* associated to the matrix A .

EXERCISE IV.1.13. Show that the projective transformations are homeomorphisms.

EXERCISE IV.1.14. Suppose we have $\alpha_i \in \mathbb{P}_{\mathbb{F}}^2$, $1 \leq i \leq 4$, none of the three of which are collinear. Show that there is a projective transformation ϕ such that $\phi(\alpha_1) = [1 : 0 : 0]$, $\phi(\alpha_2) = [0 : 1 : 0]$, $\phi(\alpha_3) = [0 : 0 : 1]$ and $\phi(\alpha_4) = [1 : 1 : 1]$.

The two dimensional projective geometry admits a duality between points and lines. Roughly speaking, the duality says that any statement about points and lines in $\mathbb{P}_{\mathbb{F}}^2$ remains true if we exchange the words “point” and “line”, and “lies on” and “contains”. The following is a manifestation of the duality in Desargues’ Theorem.

THEOREM IV.1.15. Let t_i , $i = 1, 2, 3$ be three points on a projective line $c \subset \mathbb{P}_{\mathbb{F}}^2$. Let a_i , b_i be lines containing t_i such that a_i , b_i and c are distinct. Let l_{ij} (resp. m_{ij}) be the intersection of a_i and a_j (resp. b_i and b_j). Let p_{ij} be the line containing l_{ij} and m_{ij} . Then, p_{12}, p_{13}, p_{23} have a common point.

EXERCISE IV.1.16. Prove Theorem IV.1.15.

REMARK IV.1.17. Let $G_{2,3}$ be the set of all 2-dimensional linear subspaces of $\mathbb{R}^3$. Then there is a bijection $\phi : G_{2,3} \rightarrow \mathbb{P}_{\mathbb{R}}^2$ sending a 2-dimensional linear subspace to its orthogonal complement. Therefore, $G_{2,3}$ is given a topology so that the space $G_{2,3}$ is homeomorphic to $\mathbb{P}_{\mathbb{R}}^2$.

EXERCISE IV.1.18. Let $O(3)$ be the group of 3×3 orthogonal matrices, regarded as a subspace of $\mathbb{R}^9$. Let

$$H_{1,2} = \left\{ \begin{bmatrix} \pm 1 & 0 & 0 \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & a_{33} \end{bmatrix} \mid \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} \in O(2) \right\}.$$

Then $H_{1,2}$ is a subgroup of $O(3)$. Show that the quotient set $O(3)/H_{1,2}$ is homeomorphic to $\mathbb{P}_{\mathbb{R}}^2$.

EXERCISE IV.1.19. For $0 < m < n$, generalize the construction in Exercise IV.1.18 to construct a topology on $G_{m,n}$, the set of all m -dimensional linear subspaces of $\mathbb{R}^n$, so that $G_{1,n}$ is homeomorphic to $\mathbb{P}_{\mathbb{R}}^{n-1}$, and satisfies the following property: For any $V \in G_{m,n}$ with basis $v_1, \dots, v_m$, any $w \in \mathbb{R}^n$, and any open neighborhood U of V in $G_{m,n}$, there is a $\delta > 0$ such that for any $t \in \mathbb{R}$ with $0 < t < \delta$, the linear subspace generated by $v_1 + tw, v_2, \dots, v_n$ is of dimension m and lies in U .

$$H_{i,j} = \left\{ \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \mid A \in O(i), B \in O(j) \right\}.$$

IV.2. Projection from a point, the cross-ratio

Let L, M be different lines on the Euclidean plane, and let p be a point on the plane not on L or M . Let a be a point on L such that the line $L(p, a)$ is not parallel

to M . Then $L(p, a)$ intersects M at a point which we denote by $\varphi(a)$. Therefore we have defined a map

$$\varphi : L - (?) \rightarrow M$$

where $(?)$ is either $\emptyset$ or a one-point set.

It is not hard to see that in general, φ does not preserve distance, meaning that for $a, b \in L$ such that $\varphi(a), \varphi(b)$ are defined, it is not necessarily true that $|ab| = |\varphi(a)\varphi(b)|$.

We proceed to introduce a very sophisticated quantity that is preserved by φ .

DEFINITION IV.2.1 (the cross-ratio, Euclidean). Let $v \neq 0$ be a vector in $\mathbb{R}^2$ and let $p \in \mathbb{R}^2$. Let

$$L = \{p + tv : t \in \mathbb{R}\}$$

be a line on the Euclidean plane. Let

$$a_i = p + t_i v, 1 \leq i \leq 4$$

be 4 distinct point on L . The *cross-ratio* of $a_1, \dots, a_4$ (in that order) is defined by

$$(a_1, a_2, a_3, a_4) := \frac{(t_3 - t_1)(t_4 - t_2)}{(t_4 - t_1)(t_3 - t_2)}.$$

EXERCISE IV.2.2. Show that the value of the cross-ratio is independent of the choice of p and v .

To make things nicer and more general, we work with $\mathbb{F}$ instead of $\mathbb{R}$, and with the projective plane instead of the Euclidean plane.

DEFINITION IV.2.3. Let $L \subset \mathbb{P}_{\mathbb{F}}^2$ be the projective line defined by

$$L = \{[s\xi_0 + t\eta_0 : s\xi_1 + t\eta_1 : s\xi_2 + t\eta_2] \mid s, t \in \mathbb{F}\}$$

where the notations are taken from Remark IV.1.7. Let $\omega \in \mathbb{R}^3$ not lie in the linear subspace associated to L (in the sense of Remark IV.1.7). Let $L' \in \mathbb{P}_{\mathbb{F}}^2$ be a projective line with associated linear subspace H , and let $\omega \in \mathbb{R}^3 - H$. Let $\tilde{\varphi}$ be the linear projection

$$\tilde{\varphi} : \mathbb{R}^3 = (\omega) \oplus H \rightarrow H.$$

Finally, let $p = [\omega] \in \mathbb{P}_{\mathbb{F}}^2$, and define the projection of L onto L' from p as

$$\varphi : L \rightarrow L', [x] \mapsto [\tilde{\varphi}(x)].$$

EXERCISE IV.2.4. Check that the above projection is well-defined and extends the projection from a point in the Euclidean plane.

We proceed to show that the cross ratio is preserved by any projection from a point.

PROPOSITION IV.2.5. Let $L, L' \subset \mathbb{P}_{\mathbb{F}}^2$ be projective lines and let $a_i, 1 \leq i \leq 4$ be 4 distinct point on $L \cap U_0$. Let $p \in U_0 - L - L'$. Let $\varphi : L \rightarrow L'$ be the projection from p . Then we have

$$(a_1, a_2, a_3, a_4) = (\varphi(a_1), \varphi(a_2), \varphi(a_3), \varphi(a_4)).$$

PROOF. Let the notations be as in Definition IV.2.3. Let

$$(IV.2.1) \quad a_i := [s_i \xi_0 + t_i \eta_0 : s_i \xi_1 + t_i \eta_1 : s_i \xi_2 + t_i \eta_2], \quad 1 \leq i \leq 4.$$

Let $\xi' = \tilde{\varphi}(\xi), \eta' = \tilde{\varphi}(\eta)$. Then we have

$$(IV.2.2) \quad \varphi(a_i) = [s_i \xi'_0 + t_i \eta'_0 : s_i \xi'_1 + t_i \eta'_1 : s_i \xi'_2 + t_i \eta'_2].$$

We identify $\mathbb{R}^2$ with U_0 . For $1 \leq i, j \leq 4$, $i \neq j$, we have

$$\overrightarrow{a_j a_i} = \left(\frac{s_i \xi_1 + t_i \eta_1}{s_i \xi_0 + t_i \eta_0} - \frac{s_j \xi_1 + t_j \eta_1}{s_j \xi_0 + t_j \eta_0}, \frac{s_i \xi_2 + t_i \eta_2}{s_i \xi_0 + t_i \eta_0} - \frac{s_j \xi_2 + t_j \eta_2}{s_j \xi_0 + t_j \eta_0} \right).$$

A routine computation shows

$$\overrightarrow{a_j a_i} = (s_i \xi_0 + t_i \eta_0)^{-1} (s_j \xi_0 + t_j \eta_0)^{-1} (s_i t_j - s_j t_i) (\xi_1 \eta_0 - \xi_0 \eta_1, \xi_2 \eta_0 - \xi_0 \eta_2).$$

Upon computing the cross-ratio, many of the factors cancel each other out and we have

$$(IV.2.3) \quad (a_1, a_2, a_3, a_4) = (s_1 t_3 - s_3 t_1)(s_2 t_4 - s_4 t_2)(s_1 t_4 - s_4 t_1)^{-1}(s_2 t_3 - s_3 t_2)^{-1}.$$

Comparing (IV.2.1) and (IV.2.2), we see that computing $(\varphi(a_1), \dots, \varphi(a_4))$ is simply a repetition of the above computation with ξ, η replaced by ξ', η' . However, as shown in (IV.2.3), ξ, η are canceled out. Therefore the two cross-ratios are equal. $\square$

The equation (IV.2.3) justifies the following

DEFINITION IV.2.6 (the cross-ratio, projective). Let

$$L = \{[s\xi_0 + t\eta_0, s\xi_1 + t\eta_1, s\xi_2 + t\eta_2] : s, t \in \mathbb{F}\}$$

be a projective line in $\mathbb{P}_{\mathbb{F}}^2$. Let

$$a_i = [s_i \xi_0 + t_i \eta_0, s_i \xi_1 + t_i \eta_1, s_i \xi_2 + t_i \eta_2], \quad 1 \leq i \leq 4$$

be distinct points on L . Then the *cross-ratio* of $a_1, \dots, a_4$ is

$$(a_1, a_2, a_3, a_4) = (s_1 t_3 - s_3 t_1)(s_2 t_4 - s_4 t_2)(s_1 t_4 - s_4 t_1)^{-1}(s_2 t_3 - s_3 t_2)^{-1}.$$

REMARK IV.2.7. Notice that Definition IV.2.6 makes no reference to the metric in the Euclidean plane.

EXERCISE IV.2.8. Show that the cross-ratio is invariant under projective transformations.

EXERCISE IV.2.9. Let $a_1, \dots, a_4$ and b_1 be on the same projective line and let them be pair-wise distinct except possibly a_1 and b_1 . Show that $a_1 = b_1$ if and only if

$$(a_1, a_2, a_3, a_4) = (b_1, a_2, a_3, a_4).$$

DEFINITION IV.2.10 (the cross ratio of lines). Let L_i , $1 \leq i \leq 4$ be projective lines in $\mathbb{P}_{\mathbb{F}}^2$ with a common point P . Let $M \in \mathbb{P}_{\mathbb{F}}^2$ be any projective line not containing P . Let A_i be the common point of L_i and M .

Then the *cross-ratio of the lines* is defined as

$$(L_1, L_2, L_3, L_4) := (A_1, A_2, A_3, A_4).$$

EXERCISE IV.2.11. Show that the value of (L_1, L_2, L_3, L_4) is independent of M .

Loosely speaking, any quantity that remains unchanged by projective transformations are called a projective invariant.

Projective geometry dates back to the Italian Renaissance painters, who cared much about depicting the 3-dimensional world on a piece of 2-dimensional canvas in a visually convincing fashion. For instance, as shown in Figure 1, if your eyes are the point p and you look at a painting A , which depicts the (virtual) floor B , then you would perceive a point a on A as a point $\varphi(a)$ on B , where φ is the

projection from p of A to B . Therefore to make the painting A looks perspectively convincing, it is necessary that any projective invariant derived from objects on B , such as the cross-ratio, stays the same when they are depicted on A . More on projective geometry and Renaissance art can be found in [GSS22].

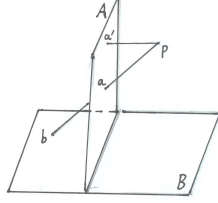


FIGURE 1. The projection of a from p is b . The point a' has no projection from p to B .

IV.3. Homogenous polynomials, projective varieties, conic sections

The study of the zero locus of a set of polynomials is one of central topics in classical and modern mathematics. To be more precise, we have the following

DEFINITION IV.3.1. Let $\{p_i\}_{i \in I}$ be a set of polynomials over $\mathbb{F}$ in variables $x_1, \dots, x_n$. The *zero locus* of $\{p_i\}_{i \in I}$ is defined as

$$V(\{p_i\}_{i \in I}) := \{(x_1, \dots, x_n) : p_i(x_1, \dots, x_n) = 0, \forall i \in I\}.$$

The zero locus of a set of polynomials is called an *affine variety*.

For instance, points, lines and quadratic curves in $\mathbb{F}^2$ are affine varieties.

A similar sort of objects can be defined in projective spaces.

DEFINITION IV.3.2. A *homogenous polynomial of degree d* over $\mathbb{F}$ in variables $x_0, \dots, x_n$ is a polynomial

$$P(x_0, \dots, x_n) = \sum_{i_0, \dots, i_n} a_{i_0, \dots, i_n} x_0^{i_0} \dots x_n^{i_n}$$

where $a_{i_0, \dots, i_n} \in \mathbb{F}$ and $i_0 + \dots + i_n = d$.

DEFINITION IV.3.3. Let $\{P_i\}_{i \in I}$ be a collection of homogenous polynomials over $\mathbb{F}$ in variables $x_0, \dots, x_n$. The *zero locus* of $\{P_i\}_{i \in I}$ is the subspace of $\mathbb{P}_{\mathbb{F}}^n$ defined by

$$V(\{P_i\}_{i \in I}) := \{[x_0, \dots, x_n] : P_i(x_0, \dots, x_n) = 0, \forall i \in I\}$$

A *projective variety* is the zero locus of a set of homogenous polynomials in $\mathbb{P}_{\mathbb{F}}^n$.

Points and projective lines in $\mathbb{P}_{\mathbb{F}}^2$ are projective varieties.

REMARK IV.3.4. For $V(\{P_i\}_{i \in I})$ to be well defined, it is essential that P_i are not arbitrary, but homogenous polynomials in $x_0, \dots, x_n$. (Why?)

Let $P(x_0, \dots, x_n)$ be a homogenous polynomial of degree d . Then there is a unique polynomial $\hat{P}(y_1, \dots, y_n)$ such that

$$P(x_0, \dots, x_n) = x_0^d \hat{P}\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

Recall the homeomorphism

$$\varphi_0 : U_0 \rightarrow \mathbb{R}^n, [x_0, \dots, x_n] \mapsto \left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

EXERCISE IV.3.5. Let $\{P_i\}_{i \in I}$ be a collection of homogenous polynomials and let V be the associated zero locus.

Show that $\varphi_0(U_0 \cap V) = V(\{\hat{P}_i\}_{i \in I})$.

Conversely, given a polynomial $p(y_1, \dots, y_n)$ of degree d , there is an associated homogenous polynomial of degree d ,

$$\bar{p}(x_0, \dots, x_n) := x_0^d p\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

DEFINITION IV.3.6. Let $V \subset \mathbb{R}^n$ be the zero locus of $\{p_i\}_{i \in I}$. The *projective completion (relative to U_0)* is the zero locus of $\{\bar{p}_i\}_{i \in I}$.

EXAMPLE IV.3.7. A projective line is the projective completion of a usual line.

DEFINITION IV.3.8. Let $Q \in \mathbb{F}[x_0, x_1, x_2]$ be a nonzero homogenous polynomial of degree 2. Then $V(Q) \in \mathbb{P}_{\mathbb{F}}^2$ is called the *conic* associated to Q .

Let $\vec{x} = (x_0, x_1, x_2)$, and let

$$Q(x_0, x_1, x_2) = \sum_i q_{ii}x_i^2 + 2 \sum_{i < j} q_{ij}x_i x_j,$$

Then we have a bilinear form $\bar{Q} = (q_{ij})$ such that $Q(x_0, x_1, x_2) = \bar{Q}(\vec{x}, \vec{x})$.

DEFINITION IV.3.9. A conic $V(Q)$ is *non-degenerate* if the bilinear form $\bar{Q}$ is non-degenerate.

The classification theorems of bilinear forms immediately yields

THEOREM IV.3.10. For $\mathbb{F} = \mathbb{C}$, there is a projective transformation from a conic S to one of the following:

- (1) $V(x_0^2 + x_1^2 + x_2^2)$;
- (2) $V(x_0^2 + x_1^2)$;
- (3) $V(x_0^2)$.

For $\mathbb{F} = \mathbb{R}$, there is a projective transformation from a conic S to one of the following:

- (1) $V(x_0^2 + x_1^2 + x_2^2) = \emptyset$;
- (2) $V(x_0^2 + x_1^2 - x_2^2)$;
- (3) $V(x_0^2 + x_1^2)$;
- (4) $V(x_0^2 - x_1^2)$;
- (5) $V(x_0^2)$.

EXERCISE IV.3.11. For $\mathbb{F} = \mathbb{R}$, show that the projective completions of

- ellipses: $a_1x_1^2 + a_2x_2^2 = 1$, $(a_1, a_2 > 0)$,
- parabolas: $ax_1^2 = y$, $a \neq 0$;
- hyperbolas: $a_1x_1^2 - a_2x_2^2 = 1$, $(a_1, a_2 > 0)$

are all projective equivalent to a non-degenerate conic.

EXERCISE IV.3.12. In Exercise IV.1.4, we have shown that the projective completion and the closure of an affine line U_0 are the same. Is this true for an arbitrary projective variety? Prove your answer.

REMARK IV.3.13. The fact revealed by Exercise IV.3.11 can be indicated by Figure 2 in which the three types of real quadratic curves are cross-sections of the conic by planes in three positions relatively different from the conic.

It is likely that the ancient Greek mathematician Menaechmus (?380-320 BC) identified the three types of real quadratic curves as conic sections, a remarkable achievement at the time.

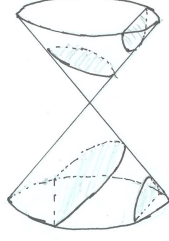


FIGURE 2. The three types of real non-degenerate quadratic curves are conic sections.

We would like to define the tangent line of a conic S at a point $p \in S$. What should it look like? Imagine there is a curve on S near P , i.e., a continuous map

$$\sigma : (-\epsilon, \epsilon) \rightarrow S, \sigma(0) = p.$$

Since $p \in U_i$ for some i , we may take ϵ small enough so that the image of σ lies in U_i , which is homeomorphic to $\mathbb{R}^2$. Therefore, we regard σ as a curve in $\mathbb{R}^2$, and we further assume it is differentiable. Now suppose $S = V(Q)$. Then we have $\bar{Q}(\sigma(t), \sigma(t)) = 0$. Taking derivative and applying the matrix product rule, we have

$$0 = \frac{d}{dt} \bar{Q}(\sigma(t), \sigma(t)) = \bar{Q}\left(\frac{d}{dt} \sigma(t), \sigma(t)\right) + \bar{Q}\left(\sigma(t), \frac{d}{dt} \sigma(t)\right) = 2\bar{Q}\left(\frac{d}{dt} \sigma(t), \sigma(t)\right),$$

which justifies the following

DEFINITION IV.3.14. Let $S = V(Q)$ be a conic, and let $p \in S$. Then the tangent line of S at p is the projective line

$$T_p(S) = \{x \in \mathbb{P}_{\mathbb{F}}^2 : \bar{Q}(x, p) = 0.\}$$

EXERCISE IV.3.15. Let L be a line and S be a non-degenerate conic, both lying in $\mathbb{P}_{\mathbb{F}}^2$.

- (1) There are at most two common points of L and S .
- (2) If $\mathbb{F} = \mathbb{C}$, then either L and S either intersect at two points, or L is a tangent line of S .

THEOREM IV.3.16 (Steiner(?)). Let $S \subset \mathbb{P}_{\mathbb{F}}^2$ be a non-degenerate conic. Let $A_i \in S$, $1 \leq i \leq 4$ be distinct points. Let $P \in \mathbb{P}_{\mathbb{F}}^2$. Then the cross-ratio

$$(PA_1, \dots, PA_4)$$

is independent of the choice of P .

If $P = A_i$, then PA_i means the tangent line of S at P .

PROOF. By Exercise IV.2.8, we may proceed after applying a projective transformation to everything.

By Exercise IV.1.14, there is a projective transformation ϕ such that $\phi(A_1) = [1, 0, 0]$, $\phi(A_2) = [0, 1, 0]$, $\phi(A_3) = [0, 0, 1]$, $\phi(A_4) = [1, 1, 1]$. Therefore, we may assume without loss of generality that $A_1 = [1, 0, 0]$, $A_2 = [0, 1, 0]$, $A_3 = [0, 0, 1]$, $A_4 = [1, 1, 1]$, and the equation of $\phi(S)$ is of the form

$$(IV.3.1) \quad (1 - k)x_1x_2 - x_1x_2 + kx_0x_2 = 0$$

for some $k \in \mathbb{F}$. Let $P = [p_0, p_1, p_2]$, and B_i , $i = 3, 4$ be the intersection of the projective line A_1A_2 and PA_i . Doing a little linear algebra, we obtain

$$B_3 = [p_0, p_1, 0], \quad B_4 = [p_0 - p_2, p_1 - p_2, 0].$$

A straightforward computation leads to

$$(PA_1, \dots, PA_4) = (A_1, A_2, B_3, B_4) = \frac{p_1p_2 - p_0p_1}{p_0p_2 - p_0p_1}.$$

Since P satisfies (IV.3.1) with x_i replaced by p_i , we have

$$(PA_1, \dots, PA_4) = \frac{p_1p_2 - p_0p_1}{p_0p_2 - p_0p_1} = \frac{(1 - k)p_0p_1 + kp_0p_2 - p_0p_1}{p_0p_2 - p_0p_1} = k$$

which does not depend on P . $\square$

THEOREM IV.3.17 (Pascal). *Let S be a non-degenerate conic, and let K be a hexagram with its vertices A_i, B_i , $1 \leq i \leq 3$ on S , with edges A_iB_j for $i \neq j$. Let $P = A_1B_2 \cap A_2B_1$, $Q = A_2B_3 \cap A_3B_2$, $R = A_1B_3 \cap A_3B_1$.*

Then P, Q, R are collinear.

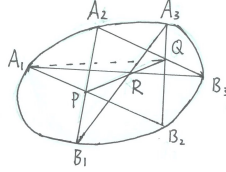


FIGURE 3.

PROOF. Let $R' = PQ \cap A_1B_3$. It suffices to show $R = R'$.

We consider the cross-ratio $(PQ, A_3B_2, A_2B_3, QA_1)$. Let $D = A_1B_2 \cap A_2B_3$. By intersecting the 4 lines with A_1B_2 , we have

$$(IV.3.2) \quad (PQ, A_3B_2, A_2B_3, QA_1) = (P, B_2, D, A_1).$$

By connecting the 4 points, P, B_2, D, A_1 to A_2 , we have

$$(IV.3.3) \quad (P, B_2, D, A_1) = (A_2B_1, A_2B_2, A_2B_3, A_2A_1).$$

By Theorem IV.3.16, we have

$$(IV.3.4) \quad (A_2B_1, A_2B_2, A_2B_3, A_2A_1) = (A_3B_1, A_3B_2, A_3B_3, A_3A_1).$$

Let $C := A_3B_2 \cap A_1B_3$. Intersecting the lines $A_3B_1, A_3B_2, A_3B_3, A_3A_1$ with A_1B_3 , we obtain

$$(IV.3.5) \quad (A_3B_1, A_3B_2, A_3B_3, A_3A_1) = (R, C, B_3, A_1).$$

Combining (IV.3.2) through (IV.3.5), we have

$$(IV.3.6) \quad (PQ, A_3B_2, A_2B_3, QA_1) = (R, C, B_3, A_1).$$

Intersecting PQ, A_3B_2, A_2B_3, QA_1 with A_1B_3 , we obtain

$$(IV.3.7) \quad (PQ, A_3B_2, A_2B_3, QA_1) = (R', C, B_3, A_1)$$

Combining (IV.3.6) and (IV.3.7), we have

$$(R, C, B_3, A_1) = (R', C, B_3, A_1)$$

It follows from Exercise IV.2.9 that we have $R = R'$. □

CHAPTER V

Differential Geometry of Regular Surfaces

V.1. Regular surfaces in $\mathbb{R}^3$

The concept of surfaces in $\mathbb{R}^3$ makes perfect sense intuitively. We proceed to define them in such a way that we can “do calculus” over them.

DEFINITION V.1.1 (Regular surfaces). A subspace $S \subset \mathbb{R}^3$ is a *regular surface* if, for each $p \in S$, there is a neighborhood V of p in S , an open set U of $\mathbb{R}^2$ and a homeomorphism

$$\begin{aligned}\phi: U &\rightarrow V \subset \mathbb{R}^3, \\ (u_1, u_2) &\mapsto (\phi_1(u_1, u_2), \phi_2(u_1, u_2), \phi_3(u_1, u_2)).\end{aligned}$$

such that ϕ_i is smooth, i.e., has continuous derivatives of all orders. The map ϕ is called a *parametrization* of S near p .

EXERCISE V.1.2. Show that a plane in $\mathbb{R}^3$ is a regular surface.

EXAMPLE V.1.3. Given $r > 0$, consider the sphere

$$S := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = r^2\}.$$

Then S is a regular sphere. Indeed, for $u_1 \in (-\pi, \pi)$, $u_2 \in (0, 2\pi)$, the function

$$\phi(u_1, u_2) = (r \cos u_1 \cos u_2, r \sin u_1 \cos u_2, r \sin u_2)$$

is a parametrization of S near any point $p \in S - \{(-1, 0, 0)\}$. A parametrization near $(-1, 0, 0)$ can be defined similarly.

EXERCISE V.1.4. Show that the following quadratic surfaces are regular surfaces:

(1) (Cylinder)

$$C = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}.$$

(2) (Hyperboloid)

$$H = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 - z^2 = 1\}.$$

(Hint: Consider the hyperbolic sine and cosine functions.)

V.2. Tangent vectors and tangent spaces

Although parametrization plays an important role in the study of regular surfaces, we notice that it is not an “intrinsic” property for regular surfaces, meaning that for a fixed regular surface S and a point $p \in S$, there may be (and always are) multiple choices of parametrization of S near p . For instance, any linear isomorphism from $\mathbb{R}^2$ to a plane $P \subset \mathbb{R}^3$ passing through 0 is a parametrization of P near 0.

One of the challenge in the study of regular surfaces is to find properties independent of the choice of parametrization. Tangent spaces is one of them.

Let S be a regular surface.

DEFINITION V.2.1. Let I be an open interval. A *smooth curve* on S is a smooth (i.e., having continuous differential of all orders) function

$$\sigma: I \rightarrow S \subset \mathbb{R}^3.$$

A smooth curve through p is a smooth curve σ such that $p \in \text{Im } \sigma$.

If $\phi: U \rightarrow V$ is a local parametrization near p , then a smooth curve through p is of the form

$$(V.2.1) \quad \sigma: I \rightarrow V, \quad t \mapsto \phi(u_1(t), u_2(t))$$

where $u_i(t)$ are smooth functions. But notice that a smooth curve on S does not depend on any parametrization.

DEFINITION V.2.2. Let $p \in S$. A *tangent vector* of S through p is the derivative of σ at t_0 , where σ is a smooth curve on S with $\sigma(t_0) = p$.

A *tangent line* of S through p is the line $p + \mathbb{R}v$ where v is a tangent vector of S through p .

DEFINITION V.2.3. Let $p \in S$. The *tangent space* of S at p , denoted by $T_p(S)$, is the set of all tangent vectors of S through p .

The *tangent plane* of S through p is the set of all points $p + v$ where v is a tangent vector.

EXERCISE V.2.4. Let $\phi: U \rightarrow V$ be a parametrization of S near p with $\phi(u^0) = p$. Show that the tangent space of S through p is the linear subspace of $\mathbb{R}^3$ spanned by $\frac{\partial \phi}{\partial u_i}(u^0)$, $i = 1, 2$. (Hint: Use (V.2.1) and the chain rule.)

V.3. The first fundamental form, length, and area

Let S be a regular surface, and let $p \in S$. Consider the tangent space $T_p(S)$ of S at p .

DEFINITION V.3.1. The *first fundamental form* of the tangent space $T_p(S)$ is the standard inner product of $\mathbb{R}^3$, regarded as a bilinear form, restricted to $T_p(S)$. It is usually denoted by I_p .

Given a local parametrization $\phi(u_1, u_2)$ of S near a point p , let

$$v^i = \frac{\partial \phi}{\partial u_i}(p), \quad i = 1, 2.$$

Then v^1, v^2 is a basis for $T_p(S)$.

With the basis v^1, v^2 , the matrix of the first fundamental form of $T_p(S)$ is

$$(V.3.1) \quad \begin{pmatrix} \langle v^1, v^1 \rangle & \langle v^1, v^2 \rangle \\ \langle v^2, v^1 \rangle & \langle v^2, v^2 \rangle \end{pmatrix}.$$

For the rest of this section, whenever there is no risk of ambiguity, we set

$$E = \langle v^1, v^1 \rangle, \quad F = \langle v^1, v^2 \rangle = \langle v^2, v^1 \rangle, \quad G = \langle v^2, v^2 \rangle.$$

EXERCISE V.3.2. Consider the sphere of radius r with local parametrization

$$(V.3.2) \quad \phi(u_1, u_2) = (r \cos u_1 \cos u_2, r \sin u_1 \cos u_2, r \sin u_2).$$

Find the first fundamental form with respect to the basis $\frac{\partial \phi}{\partial u_i}$, $i = 1, 2$.

EXERCISE V.3.3. Consider the torus parametrized by

$$(V.3.3) \quad \phi(u_1, u_2) = ((\cos u_1 + 2) \cos u_2, (\cos u_1 + 2) \sin u_2, \sin u_1).$$

Find the first fundamental form with respect to the basis $\frac{\partial \phi}{\partial u_i}$, $i = 1, 2$.

EXERCISE V.3.4. Consider the curve on the xz -plane parametrized by

$$\gamma(u_1) = (u_1^{-1}, 0, \ln(u_1 + (u_1^2 - 1)^{1/2}) - u_1^{-1}(u_1^2 - 1)^{1/2})$$

for $u_1 > 1$. Rotate γ around the z -axis to get a regular surface S (the pseudosphere). Find a local parametrization of S and the first fundamental form with respect to the basis $\frac{\partial \phi}{\partial u_i}$, $i = 1, 2$. Compare it with Example V.3.9.

The first fundamental form makes the length of a curve in S and the area of a “region” in S into “intrinsic” properties, i.e., properties determined without direct reference to the ambient space $\mathbb{R}^3$.

Let $\phi: U \rightarrow S$ be a local parametrization where U is an open set of $\mathbb{R}^2$. Let

$$\sigma = (\sigma_1, \sigma_2): I \rightarrow U$$

be a smooth function, where I is a closed interval. Then

$$\phi \cdot \sigma: I \rightarrow S$$

is a smooth curve in S .

The follow propositions are straight forward computations.

PROPOSITION V.3.5. *The length of the curve $\phi \cdot \sigma$ is*

$$(V.3.4) \quad \int_I \sqrt{E\left(\frac{d\sigma_1}{dt}\right)^2 + 2F\frac{d\sigma_1}{dt}\frac{d\sigma_2}{dt} + G\left(\frac{d\sigma_2}{dt}\right)^2} dt.$$

□

PROPOSITION V.3.6. *Suppose U is a bounded open set in $\mathbb{R}^2$. Then the area of $\phi(U)$ is*

$$(V.3.5) \quad \int_U (EG - F^2)^{1/2} du_1 du_2.$$

□

By definition, the first fundamental form is a matrix-valued function on S . Given a local parametrization ϕ from U to S , we may regard it as a function on U , which is an open set of $\mathbb{R}^2$. Notice that this function makes no reference to the way in which S is placed in $\mathbb{R}^3$. The gist of Proposition V.3.5 and Proposition V.3.6 is that, to calculate lengths and areas, it suffices to know the fundamental form. With this in mind we consider the following

DEFINITION V.3.7. Let U be an open set in $\mathbb{R}^2$. A *Riemann metric* on U is a collection of real-valued functions E, F, G on U such that for $x \in U$ the matrix

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix}$$

is positive definite.

EXAMPLE V.3.8. Let S be a regular surface with a local parametrization

$$\phi: U \rightarrow S.$$

The corresponding first fundamental form is a Riemann metric on U .

EXAMPLE V.3.9. Let $H = \{(x, y) \in \mathbb{R}^2: y > 0\}$. Let

$$E = G = y^{-2}, \quad F = 0$$

which form a Riemann metric on H . This is the *upper half-plane model for hyperbolic geometry*.

EXAMPLE V.3.10. Let $D = \{x, y \in \mathbb{R}: x^2 + y^2\}$ be the unit disk. Let

$$E = G = 4[1 - (x^2 + y^2)]^{-1}, \quad F = 0$$

which form a Riemann metric on H . This is the *Poincaré disk model for hyperbolic geometry*.

A natural question to ask is whether an arbitrary Riemann metric on U is the first fundamental form of a regular surface S with a local parametrization $\phi: U \rightarrow S$. The answer is negative. Indeed, both Riemann metrics in Example V.3.9 and Example V.3.10 are not the first fundamental form of any parametrization $\phi: H \rightarrow S$ or $\phi: D \rightarrow S$. This is a consequence of a Theorem by Hilbert (5-11, [DC16]).

V.4. Cross products in $\mathbb{R}^3$

Cross products in $\mathbb{R}^3$ is a necessary device to proceed with our investigation of regular surfaces.

DEFINITION V.4.1. Let

$$u = (u_1, u_2, u_3), \quad v = (v_1, v_2, v_3) \in \mathbb{R}^3.$$

The cross product $u \times v$ is

$$u \times v = (u_2v_3 - u_3v_2, \quad u_3v_1 - u_1v_3, \quad u_1v_2 - u_2v_1).$$

REMARK V.4.2. Let e_1, e_2, e_3 be the canonical basis of $\mathbb{R}^3$. Formally, one may write

$$u \times v = \det \begin{pmatrix} e_1 & e_2 & e_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$$

EXERCISE V.4.3. Let $\langle u, v \rangle$ denote the inner product of u and v . Let

$$w = (w_1, w_2, w_3) \in \mathbb{R}^3.$$

Show that

$$\langle w, u \times v \rangle = \det \begin{pmatrix} w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$$

EXERCISE V.4.4. Prove the following properties of cross products.

(1) Bilinearity:

$$(au + bv) \times w = au \times w + bv \times w = a(u \times w) + b(v \times w);$$

(2) Skew-symmetry: $u \times v = -v \times u$.

(3) Orthogonality:

$$\langle u \times v, u \rangle = \langle u \times v, v \rangle = 0.$$

V.5. The second fundamental form and the Gaussian curvature

Intuitively, a plane in $\mathbb{R}^3$, a sphere, and a hyperbolic surface (near the critical point) are different in the sense that they are “bent” in different ways. In this section we define the second fundamental form and the Gaussian curvature, which measures the way regular surfaces are “bent” in $\mathbb{R}^3$.

DEFINITION V.5.1. Let S be a regular surface in $\mathbb{R}^3$ and let $p \in S$. Given a local parametrization

$$\phi: U \rightarrow S$$

with $p \in \text{Im } \phi$. Let

$$\phi_i := \frac{\partial \phi}{\partial u_i}.$$

The *unit normal vector* at p is

$$\mathbf{n}(p) := \phi_1 \times \phi_2 / |\phi_1 \times \phi_2|.$$

REMARK V.5.2. Regardless of the choice of the local parametrization, the normal vector $n(p)$ is always one of the two unit vectors orthogonal to the tangent plane of p . Exactly which of them is $n(p)$ depends on the choice of the local parametrization. For the rest of this section we consider $n(p)$ only with respect to fixed local parametrization.

DEFINITION V.5.3. Let S be a regular surface in $\mathbb{R}^3$ with a local parametrization

$$\phi: U \rightarrow S.$$

Let S^2 denote the unit sphere in $\mathbb{R}^3$. The *Gaussian map*

$$g: \phi(U) \rightarrow S^2$$

is defined by $g(p) = n(p)$.

DEFINITION V.5.4. Let $v \in T_p(S)$. Let σ be a smooth path satisfying $\sigma(0) = p$ and $\sigma'(0) = v$. The *tangent map* of g , denoted by dg , sends v to $\frac{d}{dt}g(\sigma(t))|_{t=0}$.

EXERCISE V.5.5. Show that the tangent map dg is well defined linear endomorphism on $T_p(S)$. That means, proving that dg is linear, that the value of $dg(v)$ is independent of the choice of σ , and that dg takes values in $T_p(S)$.

DEFINITION V.5.6. The *second fundamental form* of S at p is the bilinear form Π on $T_p(S)$ defined by

$$\Pi(v_1, v_2) = -I(dg(v_1), v_2).$$

EXERCISE V.5.7. Let L be the matrix of Π with respect to the basis ϕ_i , $i = 1, 2$. Show that

$$L = \begin{pmatrix} \langle \mathbf{n}, \phi_{11} \rangle & \langle \mathbf{n}, \phi_{12} \rangle \\ \langle \mathbf{n}, \phi_{21} \rangle & \langle \mathbf{n}, \phi_{22} \rangle \end{pmatrix}$$

In particular, Π is symmetric.

EXERCISE V.5.8. Let T be the matrices of dg with respect to the basis ϕ_i . Let E be the matrices of I with respect to the basis ϕ_i . Prove $L = -T^t E$.

DEFINITION V.5.9. The *Gaussian curvature* of S at p is

$$K(p) := \frac{\det L(p)}{\det E(p)}.$$

THEOREM V.5.10. Let $\phi: U \rightarrow S$ be a local trivialization with $0 \in U$ and $\phi(0) = p$. Suppose $K(q) \neq 0$ in a small neighborhood of p . Let a small neighborhood Ω of 0 , let $A(\Omega)$, $A_1(\Omega)$ and $A_2(\Omega)$ denote the area of Ω , $\phi(\Omega)$ and $g\phi(\Omega)$, respectively. Then we have

$$\lim_{A(\Omega) \rightarrow 0} \frac{A_2(\Omega)}{A_1(\Omega)} = \left| \frac{\det L(p)}{\det E(p)} \right|.$$

PROOF. We have

$$A_1(\Omega) = \int_{\Omega} (\det(E))^{1/2} du_1 du_2.$$

and

$$L = -T^t E.$$

Therefore

$$A_2(\Omega) = \int_{\Omega} (\det(T^t E T))^{1/2} du_1 du_2 = \int_{\Omega} (\det(T^t E T))^{1/2} du_1 du_2 = \int_{\Omega} |\det(T)| du_1 du_2$$

and the theorem follows from the mean value theorem for integrals. $\square$

EXERCISE V.5.11. Show that the tangent map dg is self-adjoint with respect to the first fundamental form, i.e., $I(dg(v_1), v_2) = I(v_1, dg(v_2))$ for all $v_1, v_2 \in T_p(S)$. Conclude that dg has two real eigenvalues, called the *principal curvatures* of S at p , denoted by $k_1(p)$ and $k_2(p)$.

PROPOSITION V.5.12. We have

$$K(p) = k_1(p)k_2(p).$$

EXERCISE V.5.13. Prove Proposition V.5.12.

EXERCISE V.5.14. Consider the sphere of radius r with local parametrization

$$(V.5.1) \quad \phi(u_1, u_2) = (r \cos u_1 \cos u_2, r \sin u_1 \cos u_2, r \sin u_2).$$

Find the second fundamental form with respect to the basis $\frac{\partial \phi}{\partial u_i}$, $i = 1, 2$, and the Gaussian curvature.

EXERCISE V.5.15. Consider the torus parametrized by

$$(V.5.2) \quad \phi(u_1, u_2) = ((\cos u_1 + 2) \cos u_2, (\cos u_1 + 2) \sin u_2, \sin u_1).$$

Find the second fundamental form with respect to the basis $\frac{\partial \phi}{\partial u_i}$, $i = 1, 2$, and the Gaussian curvature.

EXERCISE V.5.16. Consider the pseudosphere, a regular surface parametrized by

$$(V.5.3) \quad \phi(u_1, u_2) = (u_2^{-1} \cos u_1, u_2^{-1} \sin u_1, \ln(u_2 + (u_2^2 - 1)^{1/2}) - u_2^{-1}(u_2^2 - 1)^{1/2})$$

with $u_2 > 1$. Find the second fundamental form with respect to the basis $\frac{\partial \phi}{\partial u_i}$, $i = 1, 2$, and the Gaussian curvature.

EXAMPLE V.5.17. Perhaps the simplest way to visualize the Gaussian curvature is to consider the following examples.

Let $\phi^1(u_1, u_2) = (u_1, u_2, u_1^2 + u_2^2)$ be a local parametrization of a regular surface S_1 near 0. Then the Gaussian curvature of S at 0 is positive as T has two positive eigenvalues. At 0, the two coordinate curves “bent towards the same side” of the tangent plane, which is a characteristic of positive curvature.

Let $\phi^2(u_1, u_2) = (u_1, u_2, u_1^2 - u_2^2)$ be a local parametrization of a regular surface S_2 near 0. Then the Gaussian curvature of S at 0 is negative as T has two negative eigenvalues. At 0, the two coordinate curves “bent towards different sides” of the tangent plane, which is a characteristic of negative curvature.

Let $\phi^3(u_1, u_2) = (u_1, u_2, u^2)$ be a local parametrization of a regular surface S_3 near 0. Then the Gaussian curvature of S at 0 is zero as T has two zero eigenvalues. At 0, one of the coordinate curves “does not bent” at all, which is a characteristic of zero curvature.

V.6. Gauss' Theorema Egregium

The Gaussian curvature of a regular surface depends a priori on how the surface is imbedded in $\mathbb{R}^3$. Surprisingly, however, this is true. Indeed, as shown in this section, the Gaussian curvature is determined by the first fundamental form, an “intrinsic” property of the surface. This is one of the many great observations made by Gauss.

Let S be a smooth surface in $\mathbb{R}^3$ parametrized by $\phi = \phi(u_1, u_2)$.

Let $E_{ij} = (\phi_i, \phi_j)$. Then $E = (E_{ij})$ is the first fundamental form. Let $L_{ij} = (\mathbf{n}, \phi_{ij}) = -(\mathbf{n}_i, \phi_j)$. Then $L = (L_{ij})$ is the second fundamental form. The Gaussian curvature is defined as $K = \det L / \det E$.

THEOREM V.6.1 (Gauss, Theorema Egregium). *The Gaussian curvature is determined by the first fundamental form.*

The rest of this section is occupied by the proof of Theorem V.6.1.

Defined the connection coefficients (sometimes called Christoffel symbols) $\Gamma_{ij}^k = \Gamma_{ij}^k(u_1, u_2)$ by

$$\phi_{ij} = \sum_k \Gamma_{ij}^k \phi_k + L_{ij} \mathbf{n}.$$

LEMMA V.6.2. *The functions $\Gamma_{ij}^k = \Gamma_{ij}^k(u_1, u_2)$ are determined by E and its derivatives.*

PROOF. Since $\phi_{ij} = \phi_{ji}$, we have

$$\Gamma_{ij}^k = \Gamma_{ji}^k.$$

A routine computation shows

$$\partial_k E_{ij} = \sum_l (\Gamma_{ik}^l E_{lj} + \Gamma_{jk}^l E_{il}),$$

from which we deduce

$$(\Gamma_{11}^1 \ \Gamma_{11}^2)E = (2^{-1}\partial_1 E_{11} \ \partial_1 E_{12} - 2^{-1}\partial_2 E_{11}),$$

$$(\Gamma_{21}^1 \ \Gamma_{21}^2)E = (2^{-1}\partial_2 E_{11} \ 2^{-1}\partial_1 E_{22}),$$

and

$$(\Gamma_{22}^1 \ \Gamma_{22}^2)E = (\partial_2 E_{12} - 2^{-1} \partial_1 E_2 2 \ 2^{-1} \partial_2 E_{22}).$$

The lemma then follows. $\square$

In order to perform matrix computations, we emphasize that vectors in $\mathbb{R}^3$ are regarded as column vectors. As before, let $T = (T_{ij})$ be the matrix of dg with respect to the basis ϕ_i , $i = 1, 2$. Then we have

$$(\partial_1 \mathbf{n} \ \partial_2 \mathbf{n}) = (\phi_1 \ \phi_2) T.$$

This makes sense since $(\mathbf{n}, \mathbf{n}_i) = \frac{1}{2} \partial_i (\mathbf{n}, \mathbf{n}) = 0$. A straightforward computation then shows

$$(V.6.1) \quad -L = T^t E.$$

For $i = 1, 2$, let

$$\Gamma_i = \begin{pmatrix} \Gamma_{i1}^1 & \Gamma_{i2}^1 \\ \Gamma_{i1}^2 & \Gamma_{i2}^2 \end{pmatrix}, \quad L_i = (L_{i1} \ L_{i2}), \quad T_i = \begin{pmatrix} T_{1i} \\ T_{2i} \end{pmatrix}$$

and we have

$$(V.6.2) \quad \partial_i (\phi_1 \ \phi_2 \ \mathbf{n}) = (\phi_1 \ \phi_2 \ \mathbf{n}) \begin{pmatrix} \Gamma_i & T_i \\ L_i & 0 \end{pmatrix}.$$

By $\partial_1 \partial_2 = \partial_2 \partial_1$, we have

$$\partial_1 \begin{pmatrix} \Gamma_2 & T_2 \\ L_2 & 0 \end{pmatrix} + \begin{pmatrix} \Gamma_2 & T_2 \\ L_2 & 0 \end{pmatrix} \begin{pmatrix} \Gamma_1 & T_1 \\ L_1 & 0 \end{pmatrix} = \partial_2 \begin{pmatrix} \Gamma_1 & T_1 \\ L_1 & 0 \end{pmatrix} + \begin{pmatrix} \Gamma_1 & T_1 \\ L_1 & 0 \end{pmatrix} \begin{pmatrix} \Gamma_2 & T_2 \\ L_2 & 0 \end{pmatrix}.$$

Computing the upper-left block, we have

$$\Gamma_2 \Gamma_1 + T_2 L_1 + \partial_2 \Gamma_1 = \Gamma_1 \Gamma_2 + T_1 L_2 + \partial_1 \Gamma_2,$$

from which it follows that the matrix

$$\begin{aligned} T_1 L_2 - T_2 L_1 &= (T_1 \ -T_2) \begin{pmatrix} L_2 \\ L_1 \end{pmatrix} = \begin{pmatrix} T_{11} & -T_{12} \\ T_{21} & -T_{22} \end{pmatrix} \begin{pmatrix} L_{21} & L_{22} \\ L_{11} & L_{12} \end{pmatrix} \\ &= T \begin{pmatrix} L_{21} & L_{22} \\ L_{11} & L_{12} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} L = T \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} T^t E \\ &= \begin{pmatrix} T_{12} & -T_{11} \\ T_{22} & -T_{21} \end{pmatrix} \begin{pmatrix} T_{11} & -T_{21} \\ T_{12} & -T_{22} \end{pmatrix} E = \begin{pmatrix} 0 & -K \\ K & 0 \end{pmatrix} E. \end{aligned}$$

is determined by Γ_i , $i = 1, 2$. Notice that the last equation follows from (V.6.1).

A straightforward computation shows that the $(2, 1)$ entry minus the $(1, 2)$ entry of the matrix above is $-\det T(E_{11} + E_{22})$. By Lemma V.6.2, this is $K(E_{11} + E_{22})$, and Theorema Egregium follows.

V.7. Covariant derivatives, parallel transport and geodesics

DEFINITION V.7.1. Let S be a regular surface, and let $\phi: U \rightarrow S$ be a local parametrization. A *vector field* on S is a smooth map $\nu: \phi(U) \rightarrow \mathbb{R}^3$ such that $\nu(p) \in T_{\phi(p)}(S)$ for all $p \in \phi(U)$.

Let $u: I \rightarrow U$ be a smooth curve in, and let $\sigma = \phi u: I \rightarrow \phi(U)$ be a smooth curve, $\omega = \sigma'(t_0)$, and let ν be a vector field on $\phi(U)$. Let

$$\nu(t) = \sum_{i=1}^2 \nu_i(t) \phi_i(u(t)) = \nu(\phi_i(u(t)))$$

Then a routine computation shows

$$(V.7.1) \quad \nu'(t) = \sum_{i,j,k} (\nu'_k + \nu_i u'_j \Gamma_{ij}^k) \phi_k + \sum_{i,j} \nu_i u'_j L_{ij} \mathbf{n}.$$

Inspired by the formula above, we make the following

DEFINITION V.7.2. The *covariant derivative* of ν at ω , denoted by $\nabla_\omega(\nu)$, is defined as the projection of $\nu'(t_0)$ onto $T_{\phi_i(u(t_0))}(S)$:

$$(V.7.2) \quad \nabla_\omega(\nu(t_0)) = \sum_{i,j,k} (\nu'_k + \nu_i u'_j \Gamma_{ij}^k) \phi_k(t_0).$$

REMARK V.7.3. Notice that $\nabla_\omega(\nu)$ is well defined (independent of the choice of σ) and linear in ν and ω .

A vector field ν can be defined not necessarily in an open set of S , but only along a curve $\sigma: I \rightarrow S$.

DEFINITION V.7.4. Let $\sigma: I \rightarrow S$ be a smooth curve on a regular surface S . A *vector field* ν along σ is a smooth map $\nu: \sigma(I) \rightarrow \mathbb{R}^3$ such that $\nu(t) \in T_{\sigma(t)}(S)$ for all $t \in I$.

Inspired by Definition V.7.2, we make the following

DEFINITION V.7.5. Let ν be a vector field along σ . Let

$$\nu(t) = \sum_{i=1}^2 \nu_i(t) \phi_i(u(t)) = \nu(\phi_i(u(t))).$$

The *covariant derivative* of ν at $\sigma'(t_0)$, denoted by $\nabla_{\sigma'(t_0)}(\nu)$, is defined by (V.7.2).

REMARK V.7.6. Notice the difference between Definition V.7.2 and Definition V.7.5. In the former, ν is defined in an open set of S , and the formula is a consequence of routine computation performed in $\mathbb{R}^3$. In the latter, ν is defined only along a curve, and the formula defines an “abstract” notion of covariant derivative.

The notion of parallel transport is defined in terms of covariant derivatives. Intuitively, a parallel transport of a vector ω along a curve σ is a vector field along σ which is “parallel” to ω at each point of σ , generalizing the notion of parallel vectors in $\mathbb{R}^2$.

DEFINITION V.7.7. Let $\sigma: I \rightarrow S$ be a smooth curve on a regular surface S . A *parallel transport* of a tangent vector $\omega \in T_{\sigma(t_0)}(S)$ along σ is a vector field ν along σ such that $\nabla_{\sigma'(t)}(\nu(t)) = 0$ for all $t \in I$. Two vectors in a parallel transport ν are said to be *parallel*.

EXERCISE V.7.8. Suppose for a trivialization $\phi: U \rightarrow S$, $\mathbf{I}$ is associated to the identity matrix with respect to the basis ϕ_i , $i = 1, 2$. Show that in this case, we have $\Gamma_{ij}^k = 0$ for all i, j, k . Thus, parallel transports is indeed a generalization of the notion of parallel vectors in $\mathbb{R}^2$.

EXERCISE V.7.9. Let S be the unit sphere, $0 \leq \theta < \pi/2$. Let $\sigma: I \rightarrow S$ be a smooth curve on S defined by

$$\sigma(t) = (\cos t \cos \theta, \sin t \cos \theta, \sin \theta).$$

Let ν be a vector field along σ defined by

$$\nu(t) = (-\cos t \sin \theta, -\sin t \sin \theta, \cos \theta).$$

Find the covariant derivative $\nabla_{\sigma'(0)}(\nu)$.

EXERCISE V.7.10. Let notations be as in Exercise V.7.9. Let $\theta = 0$. Show that ν is a parallel transport of $\sigma'(0)$ along σ .

PROPOSITION V.7.11. Let $\sigma: I \rightarrow S$ be a smooth curve on a regular surface S . Let $\omega \in T_{\sigma(t_0)}(S)$. Then there exists a unique parallel transport of ω along σ .

PROOF. The existence and uniqueness of parallel transports is a consequence of the existence and uniqueness of solution to the ordinary differential equation

$$(V.7.3) \quad \sum_{i,j,k} (\nu'_k + \nu_i u'_j \Gamma_{ij}^k) \phi_k(t) = 0$$

with initial condition

$$\nu(t_0) = \omega.$$

□

DEFINITION V.7.12. A *geodesic* on a regular surface S is a smooth curve $\sigma: I \rightarrow S$ such that $\sigma'(t)$ is parallel to itself along σ for all $t \in I$. In other words, $\nabla_{\sigma'(t)}(\sigma'(t)) = 0$ for all $t \in I$.

PROPOSITION V.7.13. Let $p \in S$ and $\omega \in T_p(S)$. Then there exists a unique geodesic $\sigma: [-\epsilon, \epsilon] \rightarrow S$ such that $\sigma(0) = p$ and $\sigma'(0) = \omega$ for small enough $\epsilon > 0$.

PROOF. By definition, the geodesic σ with $\sigma(0) = p$ and $\sigma'(0) = \omega$ is characterized by the ordinary differential equation (V.7.3) with the additional condition $\nu_k = u'_k$ for $k = 1, 2$:

$$\sum_{i,j,k} (u''_k + u'_i u'_j \Gamma_{ij}^k) \phi_k(t) = 0$$

with the initial condition

$$\sigma'(0) = \omega.$$

□

EXAMPLE V.7.14. Let $\phi(u, v) = (\cos u \cos v, \sin u \cos v, \sin v)$ be a parametrization of the sphere, where $0 \leq u < 2\pi$ and $-\pi/2 < v < \pi/2$. Fix v and consider the latitude curve $\sigma(t) = \phi(t, v)$. We proceed to find the parallel transport of a tangent vector

$$\nu^0 = \alpha_0 \phi_u(0, v) + \beta_0 \phi_v(0, v) \in T_{\sigma(0)}(S^2)$$

along σ and determine whether σ is a geodesic.

A routine computation shows that

$$\phi_u = (-\sin u \cos v, \cos u \cos v, 0), \quad \phi_v = (-\cos u \sin v, -\sin u \sin v, \cos v)$$

form an orthogonal basis of $T_p(S^2)$ for all $p \in \text{Im } \phi$. Therefore, the required connecting coefficients are given by

$$\Gamma_{uu}^u = \langle \phi_u, \phi_{uu} \rangle = 0, \quad \Gamma_{uu}^v = \langle \phi_u, \phi_{uv} \rangle = \sin v \cos v,$$

and

$$\Gamma_{uv}^u = \langle \phi_v, \phi_{uu} \rangle = -\sin v \cos v, \quad \Gamma_{uv}^v = \langle \phi_v, \phi_{uv} \rangle = 0.$$

Another routine computation shows that $\sigma'(t) = \phi_u(t, v)$. Suppose the parallel transport of ν^0 along σ is given by $\nu(t) = \alpha(t)\phi_u + \beta(t)\phi_v$. Then we have

$$\begin{aligned} \nabla_{\sigma'(t)}(\nu) &= \alpha' \phi_u + \alpha \Gamma_{uu}^u \phi_u + \alpha \Gamma_{uu}^v \phi_v + \beta' \phi_v + \beta \Gamma_{uv}^u \phi_u + \beta \Gamma_{uv}^v \phi_v \\ &= (\alpha' - \beta \sin v \cos v) \phi_u + (\beta' + \alpha \sin v \cos v) \phi_v. \end{aligned}$$

Therefore, ν is a parallel transport of ν_0 along σ if and only if α and β satisfy the system of ordinary differential equations

$$\begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = \begin{pmatrix} 0 & -\sin v \cos v \\ \sin v \cos v & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Solving the linear differential equation above, we have

$$\begin{pmatrix} \alpha(t) \\ \beta(t) \end{pmatrix} = \begin{pmatrix} R \cos(\sin v \cos vt + \theta) \\ R \sin(\sin v \cos vt + \theta) \end{pmatrix}$$

where R, θ are constant numbers determined by α_0 and β_0 .

REMARK V.7.15. The example above shows that the parallel transport of a tangent vector along a closed curve does not necessarily return to its original position.

EXERCISE V.7.16. Let S be the unit sphere. Show that the geodesics on S are precisely the great circles.

PROPOSITION V.7.17. Let ω, ν be two parallel transports of tangent vectors along a curve σ on a regular surface S . Then the inner product $\langle \omega(t), \nu(t) \rangle$ is constant with respect to t .

EXERCISE V.7.18. Prove Proposition V.7.17.

Another interpretation of the geodesics is given by the notion of geodesic curvature.

DEFINITION V.7.19. Let $\sigma: I \rightarrow S$ be a smooth curve on an oriented regular surface S . Let ω be a differentiable vector field along σ with unit length. Then we have

$$\nabla_{\sigma'(t)}(\omega) = \lambda(t)\mathbf{n} \times \omega$$

for some differentiable function $\lambda: I \rightarrow \mathbb{R}$. The function λ is called the *algebraic value* of σ and is denoted by $[\nabla_{\sigma'(t)}(\omega)]$.

DEFINITION V.7.20. Let $\sigma: I \rightarrow S$ be a smooth curve on a regular surface S . The curve σ is said to be *parametrized by arc length* if $I(\sigma'(s), \sigma'(s)) = 1$ for all $t \in I$.

DEFINITION V.7.21. Let $\sigma: I \rightarrow S$ be a smooth curve on an oriented regular surface S parametrized by arc length. The *geodesic curvature* of σ at s is defined as $\kappa_g(s) = [\nabla_{\sigma'(s)}(\sigma'(s))]$.

We proceed to show that the geodesics on an oriented regular surface S are precisely the curves with zero geodesic curvature.

LEMMA V.7.22. Let α, β be differentiable real-valued functions on $I = [t_0, t_1]$, such that $\alpha^2 + \beta^2 = 1$. Let $\psi_0 \in \mathbb{R}$ such that $\alpha(0) = \cos \psi_0$ and $\beta(0) = \sin \psi_0$. Then the differentiable function $\psi: I \rightarrow \mathbb{R}$ defined by

$$\psi(t) = \psi_0 + \int_0^t (\alpha\beta' - \alpha'\beta)(\tau) d\tau$$

satisfies $\alpha(t) = \cos \psi(t)$ and $\beta(t) = \sin \psi(t)$ for all $t \in I$, and $\psi(t_0) = \psi_0$.

PROOF. It suffices to show that the function

$$(\alpha - \cos \psi)^2 + (\beta - \sin \psi)^2 = 2 - 2(\alpha \cos \psi + \beta \sin \psi)$$

is constantly zero, or in other words, that

$$A := \alpha \cos \psi + \beta \sin \psi \equiv 1.$$

By $\alpha^2 + \beta^2 = 1$, we have $\alpha'\alpha + \beta'\beta = 0$. Therefore, we have $A' = 0$. Therefore, A is constant. Since $A(0) = 1$, we have $A(t) = 1$ for all $t \in I$, and the lemma follows. $\square$

LEMMA V.7.23. Let ν and ω be two differentiable vector fields along a curve $\sigma: I \rightarrow S$ on an oriented regular surface S such that $I(\nu(t), \nu(t)) = I(\omega(t), \omega(t)) = 1$ for all $t \in I$. Let ψ be a choice of angle from ν to ω . Then

$$[\nabla_{\sigma'(t)}(\omega)] - [\nabla_{\sigma'(t)}(\nu)] = \psi'$$

PROOF. Let $\bar{\nu} = \mathbf{n} \times \nu$, $\bar{\omega} = \mathbf{n} \times \omega$. Then

$$\omega = (\cos \psi)\nu + (\sin \psi)\bar{\nu},$$

$$\bar{\omega} = \nu \times \omega = (\cos \psi)\mathbf{n} \times \nu + (\sin \psi)\mathbf{n} \times \bar{\nu} = (\cos \psi)\bar{\nu} - (\sin \psi)\nu.$$

Differentiating the first equation, we have

$$\omega' = -(\sin \psi)\psi'\nu + (\cos \psi)\nu' + (\cos \psi)\psi'\bar{\nu} + (\sin \psi)\bar{\nu}'.$$

Then we have

$$\begin{aligned} \langle \omega', \bar{\omega} \rangle &= (\sin^2 \psi)\psi' + (\cos^2 \psi)\langle \nu', \nu' \rangle + (\cos^2 \psi)\psi' - (\sin^2 \psi)\langle \bar{\nu}', \nu \rangle \\ &= \psi' + (\cos^2 \psi)\langle \nu', \bar{\nu} \rangle - (\sin^2 \psi)\langle \bar{\nu}', \nu \rangle. \end{aligned}$$

Since $\langle \nu, \bar{\nu} \rangle = 0$, we have

$$\langle \nu', \bar{\nu} \rangle + \langle \nu, \bar{\nu}' \rangle = 0$$

and then

$$\langle \omega', \bar{\omega} \rangle = \psi' + \langle \nu', \bar{\nu} \rangle,$$

and the lemma follows. $\square$

COROLLARY V.7.24. The geodesics are precisely the curves with zero geodesic curvature.

EXERCISE V.7.25. Prove the corollary.

EXAMPLE V.7.26. Fix an orientation on $\mathbb{R}^2$. Let $\sigma: I = [t_0, t_1] \rightarrow \mathbb{R}^2$ be a smooth curve plane curve parametrized by arc length. Let $\omega(s) \in \mathbb{R}^2$ be a unit vector such that $(\sigma'(s), \omega(s))$ form an orthogonal basis of $\mathbb{R}^2$. Then ω is a differentiable vector field along σ .

Then the geodesic curvature of σ is simply the *curvature* of σ at $\sigma(s)$, denoted by $\kappa(s)$, characterized by

$$\sigma''(s) = \kappa(s)\omega(s).$$

Let (e_1, e_2) be the standard basis of $\mathbb{R}^2$ agreeing with the orientation. Let $\theta(s)$ be the angle from e_1 to $\sigma'(s)$. Then we have

$$\sigma'(s) = \cos \theta e_1 + \sin \theta e_2.$$

Taking derivative, we have

$$\sigma''(s) = \theta'(-\sin \theta e_1 + \cos \theta e_2).$$

Therefore, we have $\kappa(s) = \theta'(s)$, and the corollary follows. When σ is a simple closed curve, that is $\sigma(s_1) = \sigma(s_2)$ if and only if $s_1 = s_2$ or $(s_1, s_2) = (a, b)$, we have

$$\int_{\sigma} \kappa(s) = 2\pi.$$

EXERCISE V.7.27. Let S be an oriented regular surface, and let $\sigma: I = [t_0, t_1] \rightarrow S$ be a smooth simple closed curve. Let ν be a parallel vector field along σ . Let θ be the angle from $\nu(t_0)$ to $\nu(t_1)$. Let $\kappa_g(t)$ be the geodesic curvature of σ at t . Show that

$$\int_{\sigma} \kappa_g = 2\pi - \theta.$$

V.8. Euler characteristics and the Gauss-Bonnet theorem

In this section we state, without proof, the Gauss-Bonnet theorem that associates the Gaussian curvature of a regular surface (without boundary) to its global topological properties.

Let $\phi: U \rightarrow S$ be a local parametrization of an oriented regular surface S such that U is an open ball.

Let $\alpha: I \rightarrow S$ be a piecewise smooth curve on an oriented regular surface S , corresponding to the partition $s_0 < s_1 < \cdots < s_{k+1}$ of I , over each piece parametrized by arc length. Let

$$\alpha_i :: [s_i, s_{i+1}] \rightarrow S$$

be the smooth pieces of α . Let θ_i be the angle from $\alpha_i(s_{i+1})$ to $\alpha_{i+1}(s_{i+1})$.

THEOREM V.8.1. *If $s_0 = s_{k+1}$ and R is the region with boundary α in S , then we have*

$$\sum_{i=0}^k \int_{s_i}^{s_{i+1}} \kappa_g(s) ds + \int \int_R K dr + \sum_i \theta_i = 2\pi.$$

Recall that we have defined the Euler characteristics of a polyhedron as $V - E + F$ where V , E , F denotes the number of vertices, edges, and faces, respectively.

DEFINITION V.8.2. The *Euler characteristics* of a regular surface S is the Euler characteristics of a polyhedron which is homeomorphic to S . It is denoted by $\chi(S)$.

REMARK V.8.3. It is a nontrivial fact that this definition is intrinsic to S , but we will not prove it.

EXERCISE V.8.4. Find the Euler characteristics of the sphere and the circle.

THEOREM V.8.5. *Let S be a compact regular surface without boundary. Then we have*

$$\frac{1}{2\pi} \int_S K d\sigma = \chi(S),$$

where, over a local parametrization, $d\sigma = (\det E)^{1/2} du_1 du_2$.

CHAPTER VI

Hyperbolic Geometry

The main reference of this chapter is $[\mathbf{A}^+]$.

VI.1. The Riemann sphere and the complex projective plane

EXERCISE VI.1.1. Show that the equation of a line in $\mathbb{C}$ is of the form

$$(wz + \bar{w}\bar{z}) + c = 0$$

for some $w \in \mathbb{C}$ and $c \in \mathbb{R}$. (Hint: Suppose that $\bar{w}$, as a planar vector, is perpendicular to the line.)

EXERCISE VI.1.2. Show that the equation of a circle in $\mathbb{C}$ is of the form

$$z\bar{z} + (wz + \bar{w}\bar{z}) + c = 0$$

for $w \in \mathbb{C}$ and $c \in \mathbb{R}$. Can a circle defined as above be an empty set?

Consider the *Stereographic projection*

$$(VI.1.1) \quad \phi: S := S^2 - \{(0, 0, 1)\} \rightarrow \mathbb{C}, \quad (u, v, w) \mapsto \frac{u}{1-w} + i \frac{v}{1-w}$$

which is a homeomorphism. Conversely, there is a way to “adjoin” a point to $\mathbb{C}$ and recover the sphere S^2 .

DEFINITION VI.1.3. The *one point compactification* of $\mathbb{C}$ is the topological space whose underlying set is $\mathbb{C} \cup \{\infty\}$, with the topology generated by

- the open sets of $\mathbb{C}$, and
- the sets of the form $\{\infty\} \cup W$, where W is the complement of a compact set in $\mathbb{C}$.

The one point compactification of $\mathbb{C}$ is denoted by $\bar{\mathbb{C}}$.

EXERCISE VI.1.4. Show that ϕ in (VI.1.1) extends to a homeomorphism

$$\bar{\phi}: S^2 \rightarrow \bar{\mathbb{C}}.$$

The choice of the notation ∞ is justified by the fact

$$\lim_{x \rightarrow \{(0,0,1)\}} \phi(x) = \infty.$$

EXERCISE VI.1.5. Show that the function

$$f: \mathbb{C} - \{0\} \rightarrow \mathbb{C}, \quad z \mapsto z^{-1}$$

extends uniquely to a homeomorphism from $\bar{\mathbb{C}}$ to itself. Find $f(0)$ and $f(\infty)$. Furthermore, show that if L is a line in $\mathbb{C}$ not passing 0, then $f(L)$ is a circle.

REMARK VI.1.6. More generally, for polynomials f and g and $z \in \mathbb{C}$ such that $f(z) \neq 0$ and $g(z) = 0$, the function f/g extends to z by $(f/g)(z) = \infty$.

In view of Exercise VI.1.1, Exercise VI.1.10, and Exercise VI.1.5, one may regard lines and circles as unified geometric objects with equations of the form

$$(VI.1.2) \quad az\bar{z} + (wz + \bar{w}\bar{z}) + c = 0$$

for $w \in \mathbb{C}$ and $a, c \in \mathbb{R}$. We impose the extra rule that lines contain the point ∞ , whereas circles don't.

We present another point of view on this unification of lines and circles. Recall that the complex projective space $\mathbb{CP}^1$, which is defined by

$$\mathbb{CP}^1 = \{(z_1, z_2) \in \mathbb{C} \times \mathbb{C} \mid (z_1, z_2) \neq (0, 0)\} / \{(z_1, z_2) \simeq (kz_1, kz_2)\}.$$

An element in $\mathbb{CP}^1$ represented by (z_1, z_2) is denoted by $[z_1 : z_2]$.

PROPOSITION VI.1.7. *The map*

$$\psi: \bar{\mathbb{C}} \rightarrow \mathbb{CP}^1, z \mapsto \begin{cases} [z : 1], & z \in \mathbb{C}, \\ [1 : 0], & z = \infty \end{cases}$$

is a homeomorphism. Its inverse is given by

$$\psi^{-1}: \mathbb{CP}^1 \rightarrow \bar{\mathbb{C}}, [z_1, z_2] \mapsto \begin{cases} z_1/z_2, & z_2 \neq 0, \\ \infty, & z_2 = 0. \end{cases}$$

EXERCISE VI.1.8. Prove Proposition VI.1.7.

DEFINITION VI.1.9. A *circle* in $\mathbb{CP}^1$ is a subspace determined by the equation

$$az_1\bar{z}_1 + (wz_1\bar{z}_2 + \bar{w}\bar{z}_1z_2) + cz_2\bar{z}_2 = 0$$

for $w \in \mathbb{C}$ and $a, c \in \mathbb{R}$, a, c, w not all 0.

EXERCISE VI.1.10. Let ψ be as in Proposition VI.1.7. Show that the image of a circle or a line under ψ is a circle.

PROPOSITION VI.1.11. *Let z_i , $1 \leq i \leq 3$ be distinct points of $\bar{\mathbb{C}}$. Show that there is a unique circle C in $\bar{\mathbb{C}}$ such that $z_i \in C$ for all i .*

PROOF. If none of the 3 points is ∞ , then planar geometry tells us that they uniquely determine a point of $\mathbb{C}$. Otherwise, say $z_1 = \infty$, then the circle is the closure of the line passing z_2 and z_3 . $\square$

VI.2. The Möbius transformations

Let $\underline{z} = (z_1, z_2) \in \mathbb{C}^2$ be regarded as a column vector. Let $[\underline{z}]$ denote the point in $\mathbb{CP}^1$ represented by $\underline{z}$. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{C}^{2 \times 2}.$$

Let $[A\underline{z}]$ denote the point in $\mathbb{CP}^1$ represented by

$$[az_1 + bz_2 : cz_1 + dz_2].$$

DEFINITION VI.2.1. A *Möbius transformation* on $\mathbb{CP}^1$ is a homeomorphism on $\mathbb{CP}^1$ of the form

$$[\underline{z}] \mapsto [A\underline{z}]$$

where A is an invertible matrix. A *generalized Möbius transformation* is either a Möbius transformation or of the form

$$[\underline{z}] \mapsto [A\underline{\bar{z}}]$$

where A is an invertible matrix.

PROPOSITION VI.2.2. *Let m be a generalized Möbius transformation, and let C be a circle in $\mathbb{CP}^1$. Then $m(C)$ is a circle.*

PROOF. This is a routine computation. $\square$

PROPOSITION VI.2.3. *The generalized Möbius transformations form a group under compositions, such that the Möbius transformations form a subgroup.*

PROOF. The proof is a routine inspection of definitions. $\square$

The group of generalized Möbius transformations is called the *generalized Möbius group* and denoted by $\mathcal{M}$, and its subgroup of *Möbius transformations* is called the *Möbius group* and denoted by $\mathcal{M}^+$.

The definitions of (generalized) Möbius transformations gives rise to actions of $\mathcal{M}$ and $\mathcal{M}^+$ on $\mathbb{CP}^1$.

EXERCISE VI.2.4. The action of $\mathcal{M}^+$ on $\mathbb{CP}^1$ (and therefore of $\mathcal{M}$) is transitive.

Indeed, even more is true:

PROPOSITION VI.2.5. *Let $[\alpha_1 : \alpha_2]$, $[\beta_1 : \beta_2]$, $[\gamma_1 : \gamma_2]$ and $[\alpha'_1 : \alpha'_2]$, $[\beta'_1 : \beta'_2]$, $[\gamma'_1 : \gamma'_2]$ be two sets of distinct points in $\mathbb{CP}^1$. Then there is a unique $m \in \mathcal{M}^+$ satisfying*

$$m([\alpha_1 : \alpha_2]) = [\alpha'_1 : \alpha'_2], \dots$$

PROOF. It suffices to consider the special case

$$[\alpha'_1 : \alpha'_2] = [1, 0], [\beta'_1 : \beta'_2] = [1, 1], [\gamma'_1 : \gamma'_2] = [0, 1].$$

Then we let $m[\underline{z}] = [A\underline{z}]$ for

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

A simple calculation shows that there is a unique m as required, determined by

$$a = \gamma_2, b = -\gamma_1, c = \frac{\gamma_2\beta_1 - \gamma_1\beta_2}{\alpha_2\beta_1 - \alpha_1\beta_2}\alpha_2, d = -\frac{\gamma_2\beta_1 - \gamma_1\beta_2}{\alpha_2\beta_1 - \alpha_1\beta_2}\alpha_1.$$

The uniqueness of m is left as an exercise. $\square$

EXERCISE VI.2.6. Complete the proof of the uniqueness of m in the proof of Proposition VI.2.5.

Another way to define the Möbius transformations is to work with $\bar{\mathbb{C}}$.

EXERCISE VI.2.7. Show that the Möbius transformations on $\bar{\mathbb{C}}$ are maps of the form

$$z \mapsto \begin{cases} \frac{az+b}{cz+d}, & z \in \mathbb{C}, \\ \frac{a}{c}, & z = \infty, \end{cases}$$

and the generalized Möbius transformations are either the Möbius transformations or the maps of the form

$$z \mapsto \begin{cases} \frac{a\bar{z}+b}{c\bar{z}+d}, & z \in \mathbb{C}, \\ \frac{a}{c}, & z = \infty, \end{cases}$$

where in both cases the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is invertible.

REMARK VI.2.8. The above presentation of the (generalized) Möbius transformations has the advantage of being more concise. We will use it primarily for the rest of this chapter.

EXERCISE VI.2.9. Show that the unique Möbius transformation m satisfying $m(z_1) = 0$, $m(z_2) = 1$ and $m(z_3) = \infty$ is of the form

$$m(z) = \frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1}.$$

VI.3. The cross ratio

DEFINITION VI.3.1. Let z_i , $1 \leq i \leq 4$ be distinct points in $\bar{\mathbb{C}}$. The *cross ratio* of $z_1, \dots, z_4$ is

$$(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_4)(z_3 - z_2)}{(z_1 - z_2)(z_3 - z_4)}.$$

If one of the z_i 's is ∞ , take the corresponding limit.

PROPOSITION VI.3.2. For $m \in \mathcal{M}^+$ and $z_i \in \bar{\mathbb{C}}$, $1 \leq i \leq 4$, we have

$$(z_1, z_2, z_3, z_4) = (m(z_1), m(z_2), m(z_3), m(z_4)).$$

PROOF. This is a routine computation. $\square$

COROLLARY VI.3.3. For $m \in \mathcal{M} - \mathcal{M}^+$, and z_i as above, we have

$$\overline{(z_1, z_2, z_3, z_4)} = (m(z_1), m(z_2), m(z_3), m(z_4)).$$

PROPOSITION VI.3.4. Let z_i be as above. Then the z_i 's are cyclic quadrilateral (i.e., on the same circle) if and only if

$$(z_1, z_2, z_3, z_4) \in \mathbb{R}.$$

PROOF. By Proposition VI.3.2, it suffices to consider the case $z_2 = \infty$, $z_3 = 1$, $z_4 = 0$. Then we have

$$\lim_{z_2 \rightarrow \infty} \frac{z_1(1 - z_2)}{z_1 - z_2} = z_1$$

and we conclude. $\square$

COROLLARY VI.3.5. Let $m \in \mathcal{M}$ and let C be a circle in $\bar{\mathbb{C}}$. Then $m(C)$ is a circle.

EXERCISE VI.3.6. Let $m \in \mathcal{M}$. Show that $m(z) \in \bar{\mathbb{R}}$ for all $z \in \bar{\mathbb{R}}$ if and only if there are $a, b, c, d \in \mathbb{R}$, $ad - bc \neq 0$, such that m is of the form

$$m(z) = \frac{az + b}{cz + d}$$

or

$$m(z) = \frac{a\bar{z} + b}{c\bar{z} + d}.$$

By Proposition VI.3.4, if a circle C contains three distinct points z_i , $1 \leq i \leq 3$, then C divides $\bar{\mathbb{C}}$ into two disjoint open sets

$$\{z \in \bar{\mathbb{C}} \mid (z, z_1, z_2, z_3) > 0\}$$

and

$$\{z \in \bar{\mathbb{C}} \mid (z, z_1, z_2, z_3) < 0\}.$$

We call each of them an (open) disk.

COROLLARY VI.3.7. *Let D_i , $i = 1, 2$, be a disk with boundaries ∂D_i . If $m \in \mathcal{M}$ satisfies $m(D_1) = D_2$, then $m(\partial D_1) = \partial D_2$.*

VI.4. The conformality of elements of $\mathcal{M}^+$

Let m be a Möbius transformation defined by

$$m(z) = \frac{az + b}{cz + d}$$

for $ad - bc \neq 0$. Regarding m as a smooth function from $\mathbb{R}^2 - \{-d/c\}$ (regarding the complex number $-d/c$ as a vector in $\mathbb{R}^2$) to $\mathbb{R}^2$, the derivative of m is 2×2 real matrix

$$m' = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}$$

where $m(x + iy) = u + iv$.

PROPOSITION VI.4.1 (Cauchy-Riemann formula). *The matrix m' satisfies*

$$u_x = v_y, u_y = -v_x,$$

or equivalently

$$m' = r \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

i.e., m' is a rotation matrix.

PROOF. The proof is a routine computation. □

DEFINITION VI.4.2. A *conformal map* from an open set U of $\bar{\mathbb{C}}$ to $\bar{\mathbb{C}}$ is a smooth map f such that for two smooth path ϕ and ψ in U intersecting at $p \in U$, the angle between the tangent vectors of ϕ and ψ at p is equal to the angle between the tangent vectors of $f \cdot \phi$ and $f \cdot \psi$ at $f(p)$.

If $p = \infty$, then replace ϕ and ψ with $f \cdot \phi$ and $f \cdot \psi$, respectively, and replace p with 0. similarly if $f(p) = \infty$, then replace f is $1/f$.

PROPOSITION VI.4.3. *The Möbius transformation is a conformal map.*

PROOF. When ∞ is not involved, this is a consequence of Proposition VI.4.1. To deal with ∞ , see Exercise VI.4.4. □

EXERCISE VI.4.4. Let

$$\sigma, \tau: (-\epsilon, \epsilon) \rightarrow \bar{\mathbb{C}}, \quad \sigma(0) = \tau(0) = \infty.$$

How would you define the angle from $\frac{d\sigma}{dt}(0)$ and $\frac{d\tau}{dt}(0)$ in such a way that Möbius transformations are conformal?

EXERCISE VI.4.5. Consider the curves

$$\sigma, \tau: (-1, 1) \rightarrow \bar{\mathbb{C}}, \quad \sigma(t) = \frac{1}{t}, \quad \tau(t) = \frac{t+i}{t}.$$

Find the cosine of the angle from $\frac{d\sigma}{dt}(0)$ to $\frac{d\tau}{dt}(0)$.

EXERCISE VI.4.6. If $m \in \mathcal{M} - \mathcal{M}^+$, then m inverts the angles between tangent lines.

VI.5. The upper half plane model of hyperbolic geometry

In this section we establish the basics of hyperbolic geometry, in terms of what is called the “upper half plane” model. Let

$$\mathbb{H} := \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}$$

be the upper half plane.

DEFINITION VI.5.1. A (*hyperbolic*) *line* in $\mathbb{H}$ is either a half line perpendicular to $\mathbb{R}$, or a half circle of which the center lies in $\mathbb{R}$.

Two lines in $\mathbb{H}$ are *parallel* if they are disjoint.

PROPOSITION VI.5.2. *For any distinct points α, β on L , there is a unique line L passing through α, β .*

The proof is elementary and straightforward.

PROPOSITION VI.5.3. *Let L be a line in $\mathbb{H}$. Let $p \in \mathbb{H}$. Then there are infinitely many lines containing p parallel to L .*

EXERCISE VI.5.4. Prove Proposition VI.5.3.

We consider the (generalized) Möbius transformations that maps $\mathbb{H}$ to itself. Let

$$\mathcal{M}_{\mathbb{H}} := \{m \in \mathcal{M} \mid m(\mathbb{H}) = \mathbb{H}\},$$

and let

$$\mathcal{M}_{\mathbb{H}}^+ := \{m \in \mathcal{M}^+ \mid m(\mathbb{H}) = \mathbb{H}\}.$$

PROPOSITION VI.5.5. *The elements of $\mathcal{M}_{\mathbb{H}}^+$ are elements m of the form*

$$m(z) = \frac{az + b}{cz + d}$$

for $a, b, c, d \in \mathbb{R}$ such that $ad - bc > 0$. The elements of $\mathcal{M}_{\mathbb{H}}$ are either in $\mathcal{M}_{\mathbb{H}}^+$ or elements m of the form

$$m(z) = \frac{a\bar{z} + b}{c\bar{z} + d}$$

for $a, b, c, d \in \mathbb{R}$ such that $ad - bc < 0$.

PROOF. By Corollary VI.3.7, we have $\mathcal{M}_{\mathbb{H}} \subset \mathcal{M}_{\mathbb{R}}$ and $\mathcal{M}_{\mathbb{H}}^+ \subset \mathcal{M}_{\mathbb{R}}^+$. The rest is a routine computation. $\square$

PROPOSITION VI.5.6. *The elements of $\mathcal{M}_{\mathbb{H}}$ takes hyperbolic lines to hyperbolic lines.*

PROOF. This follows from Proposition VI.4.3, that the Möbius transformations are conformal. $\square$

PROPOSITION VI.5.7. *The group $\mathcal{M}_{\mathbb{H}}$ acts transitively on $\mathbb{H}$.*

VI.6. Length and Distance in $\mathbb{H}$

Throughout this section, I denotes a closed interval $[\alpha, \beta]$.

DEFINITION VI.6.1. Given a smooth curve $\sigma: I \rightarrow \mathbb{H}$, the *hyperbolic length*, or simply length, if there is no risk of ambiguity, is the integral

$$\text{length}(\sigma) = \int_{\alpha}^{\beta} (\text{Im } \sigma(t))^{-1} |\sigma'(t)| dt.$$

A more convenient convention is

$$\text{length}(\sigma) = \int (\text{Im } z)^{-1} |dz|$$

where we denote $\sigma(t)$ by z .

PROPOSITION VI.6.2. *The length of a smooth curve is invariant under generalized Möbius transformations. More precisely, letting $m \in \mathcal{M}_{\mathbb{H}}^+$ and letting σ be a smooth curve, we have*

$$\text{length}(\sigma) = \text{length}(m(\sigma)).$$

PROOF. We consider the case $m \in \mathcal{M}_{\mathbb{H}}$. The general case is left as an exercise. (In fact, try to complete the whole proof by yourself!)

Suppose $m(z) = \frac{az+b}{cz+d}$. We have

$$(VI.6.1) \quad |dm(z)| = \left| d \frac{az+b}{cz+d} \right| = \left| \frac{ad-bc}{(cz+d)^2} dz \right|,$$

and

$$(VI.6.2) \quad \text{Im}(mz) = \text{Im} \frac{az+b}{cz+d} = |cz+d|^{-2} (ad-bc) \text{Im } z.$$

The desired result follows from (VI.6.1) and (VI.6.2). $\square$

REMARK VI.6.3. One may compare the role of the Euclidean group $SE(2)$ generated by rotations, reflections and parallel translations with the generalized Möbius group $\mathcal{M}_{\mathbb{H}}$. They are indeed both isometry groups, but with respect to different geometries: Euclidean geometry and hyperbolic geometry.

In Euclidean geometry, the length of the line segment joining two given points p and q the least among the lengths of all smooth curves joining p and q . Moreover, the length of the line segment between p and p defines the metric in the Euclidean plane. Similar facts hold in hyperbolic geometry. Recall Proposition VI.5.2, which states that for each two points in $\mathbb{H}$, there is a unique line joining them.

We present the following theorem without a proof.

THEOREM VI.6.4. *Let $p, q \in \mathbb{H}$. Let σ be the unique line segment joining p, q . Let $\delta(p, q)$ be the infimum of the length of all curves joining P and q . Then we have*

$$\delta(p, q) = \text{length}(\sigma).$$

Furthermore, δ defines a metric over $\mathbb{H}$, making $\mathbb{H}$ a metric space.

For a detailed discussion of the aforementioned metric, see § 3.3 and § 3.4 of $[\mathbf{A}^+]$.

REMARK VI.6.5. In Chapter V, Exercise V.3.4 and Exercise V.5.16, we consider the pseudosphere, a regular surface parametrized by

$$(VI.6.3) \quad \phi(u_1, u_2) = (u_2^{-1} \cos u_1, u_2^{-1} \sin u_1, \ln(u_2 + (u_2^2 - 1)^{1/2}) - u_2^{-1}(u_2^2 - 1)^{1/2})$$

with $u_2 > 1$. The first fundamental form with respect to this basis is

$$(VI.6.4) \quad \frac{du_1^2 + du_2^2}{u_2^2}$$

which is compatible with the line element $\Im z^{-1} dz$ that gives us the hyperbolic lengths, for $u_2 > 1$.

Therefore, the “intrinsic” geometry of $\mathbb{H}$, at least of its subspace $\{\operatorname{Im} z > 1\}$, is the “same geometry ” as that of the pseudosphere. By Exercise V.5.16 and Theorem V.6.1, we say that hyperbolic geometry is of negative curvature, although we do not “see” how $\mathbb{H}$ is bent.

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